IBU biathlon World Cup points and prize money distribution analysis

2nd edition

Juho Toivonen

March 2026

Abstract

This report analyzes the IBU biathlon World Cup points and prize money distribution and compares the FIS cross-country World Cup prize money distribution to the respective IBU version. The analysis tells that points distribution is fairly good and IBU is doing better job on distributing money even with the advantage of having more to share. In both aspects there are still room for improvement.

Contents

1 Introduction

Cross-country skiing is a way of movement in the snow with skis and poles. It's separated from alpine skiing where the skis are used to only come down the hills or mountains. It's origins are in historical times with people who have been living in the areas of snow cover during the periodic cycle. These people have been using wooden skis to move in snowy terrain [11]. In current day cross-country skiing is seen as popular winter activity especially in Nordic countries but it's also a competitive sport. It's been part of the Olympic Winter Games from 1924 in Chamonix [12].

Biathlon is a winter sport which combines cross-country skiing and shooting. It's roots are in Scandinavian survival methods but it has slowly grown as a spectator sport through military competitions. It was part of Olympic Winter Games in 1924 in Chamonix as a Military Patrol and slowly started evolving towards the sport which it's now after 1950s [4].

In the highest international level, cross-country competitions are part of International Ski and Snowboard Federation (short FIS). It includes Winter Olympic Games, FIS Nordic World SKi Championships, FIS Cross-Country World Cup, Tour de Ski, Ski Classics series and more [1]. Basically FIS controls all the major skiing sport events apart from biathlon.

In current day biathlon has it's World Cup tour which is running every year and World Championships which are held on every year other than Winter Olympic years. International Biathlon Union (short IBU) is in charge of the yearly schedule of IBU World Cup and IBU World Championships and they also decide the competition locations and formats. IBU also organizes IBU Cup which are meant is "tier 2" international competition circuit underneath IBU World Cup and IBU Junior Cup which provides international circuit for youth athletes. In the IBUs yearly calendar there are also IBU Open European Championships which is the highlight event of year for the IBU Cup circuit, IBU Youth Junior World Championships and IBU Junior Open European Championships for young athletes and they also provide Para Biathlon events for disabled athletes [6].

In this analysis the main focus will be on events under the umbrella of IBU. It includes analysis and comparison of the point distributions used in both competitions throughout the year, comparison of distributions between the genders and comparison of prize money distribution between IBU ja FIS World Cups. The goal is to take a deeper dive behind the results and also discuss the prize money distribution and it's healthiness towards the athletes competing in the sport.

2 Competition formats

2.1 IBU World Cup

Over the years the competition formats have changed remarkably but for the last years the format has been fairly constant. The season 2025-2026 consists of the following race formats:

Sprint (7 races)
Pursuit (7 races)
Individual (3 races)
Mass (4 races)

Relay (5 races)
Mixed Relay (3 races)
Single Mixed Relay (3 races)

[9]

The normal format for the season is that there are three (3) competition weekends in a row in three (3) locations and after that there is a longer break. These are called trimesters [10]. Between second and third trimester on each year there is either IBU World Championships or Olympic Winter Games.

For the each competition weekend there are usually three (3) competition days for each gender which are distributed into 3-5 days depending on the weekend. All the competitions use skating style of skiing. Mixed relays are competed always in the same day and usually athletes will be competing in one (mixed or single mixed relay) race but for the smaller countries some athletes might be participating on both. Sprint, pursuit, individual and mass disciplines are counted towards Total Score (explained later) and sprint, individual and relays have separate point system which are counted towards Nations Cup. Nations Cup is not touched in this analysis but it's important for the countries and it determines how many athletes each country is eligible at naming for each competition for the next season IBU World Cup. There are also prize money awarded in each competition [6].

IBU World Cup in the season 2025-2026 contains total of 21 individual competition and 11 relays [10]. The following chapters will explain separate competition formats more closely based on information in [5].

2.1.1 Sprint

Sprint competition is the shortest of the individual competitions in the circuit. Women will ski total of 7,5 kilometers and men total of 10 kilometers. The race contains two (2) shootings, one (1) prone, one (1) standing, and equal amount of skiing in between of the shootings. The athlete will try to shoot five (5) targets with five (5) shots on each shooting and for every miss they have, they have to ski extra 150 meter loop right after the shooting before continuing for the next lap of skiing. The winner is the fastest to the finish line and the results of the sprint competitions determines start times for the pursuit competition.

Sprint competition can be seen as the most vital competition form of the current circuit. It is competed the most of the individual competitions, all the eligible athletes are open to participate on it and the results will have significant effect on the next competition, the pursuit, which is always competed in the same weekend with the sprint competition. It also counts towards the Nations Cup. Usually sprints had over 100 athletes participating on each time it was competed in the season 2025-2026 for both genders.

2.1.2 Pursuit

Pursuit competition is usually competed as a pair with sprint as the sprint competition determines the starting intervals for the pursuit competition. Only the top 60 athletes of the previous sprint competition are allowed to participate in the pursuit. In the pursuit competition women ski total of 10 kilometers and men ski 12,5 kilometers. It contains four (4) shootings, two (2) prone, two (2) standing and equal amount of skiing in between the shootings. The each individual shooting work as the same as in sprint. For every miss the athlete must ski extra 150 meter loop before continuing for the next skiing lap. The winner is the first to finish the competition.

In season 2025-2026 the pursuit competition followed the sprint every single time and you could see them as pairs of competitions. In earlier years pursuit has been sometimes competed also after individual competition where the results individual have been the factor of the starting intervals although the time differences have been halved. It can be more tactical format from strategy point of view and also the effect of missing the shots are much more visible during the competition. It's also very dependent on athletes capability of competing in sprint because bad results in sprints will almost quarantee to losing the opportunity to do well in the pursuits. But there are big jumps and big losses of positions seen on every pursuit which means that almost everything is still possible.

Pursuit doesn't count towards Nations Cup.

2.1.3 Individual

Individual competition is the closest to the original military formats as it's the longest competition from them all. Women ski total of 15 kilometers and men total of 20 kilometers. Once again the race contains four (4) shootings. For the individual race the shooting style alternates in between of the shooting so the order is prone, standing, prone, standings. Each athlete is still shooting five (5) targets with five (5) shots. For every miss athlete will get a penalty minute for their overall time. The winner is the athlete who finished the race in fastest time.

This competition format can be seen as the original format and the name also suggest it being a single individual competition in the history. It's space in the calendar has been diminished for quite some time and some years the IBU World Cup has contained only two (2) individual races for the whole year. For the season 2025-2026 it was raced three (3) times although one of the races was shortened version where skiing distance was lowered and penalty for the miss was only 45 seconds. It's the other individual competition which is counted towards the Nations Cup and countries can place the athletes in the competition which are eligible. For this reason this competition also averaged around 100 athletes per competition in season 2025-2026.

2.1.4 Mass

Mass competition is final individual competition form. Only 30 athletes can participate into this competition format at one time. In the weekends when it's raced, the top 25 of Total Score are eligible to participate and 5 of the best athletes outside of top 25 registers their eligibility by their performances during each weekend. There is also a reserve list to fill up the spots which might not be filled to make sure that the competition is always ran with 30 athletes.

Women ski 12,5 kilometers and men 15 kilometers. This format also includes four (4) shootings, two (2) prone, two (2) standing. Individual shootings are also similar and for each miss the athlete must ski the 150 meter extra loop. The winner is the first to the finish line.

This race can be seen as the ultimate goal for the athetes to fill their competition calendar as it contains the smallest amount and the best athletes at each given time. Since it covers the best athletes starting from at the same time, it makes the competitions very exciting and tight until to the end. This competition doesn't count towards the Nations Cup.

2.1.5 Relays

IBU World Cup has integrated relay competitions tightly into their schedule as there are total of 8 relay days in 9 weekends. It's important for countries to participate to the relays as all the relay competitions counts towards Nations Cup. For this reason in each competition there is an average of 20 countries participating. None of the countries are allowed to set two teams into the relays.

Men's/Women's/Mixed relays work in same way. There are four (4) legs with one (1) athlete competing on each leg for the team. The skiing distance varies but there's always two (2) shootings, one (1) prone and one (1) standing. Specialty in shooting on relays are that each athlete are allowed three (3) extra bullets per shooting in case they miss any of their first five (5) shots. If after eight (8) shots they still haven't hit all the five (5) targets, they have to ski the extra 150 meter loop for each missed target left. In mixed relays the starting order changes for each competition. For every other competition women's legs will be competed first and men will finish the competition and the next one the positions will switch.

Single mixed relay is the newest edition but it's still been competed at least on last six (6) seasons and it has taken it's place in the same day with mixed relay. In single mixed relay one (1) woman and one (1) man athlete will do two legs each. Each leg contains short distance of skiing and two (2) shootings, one (1) prone, one (1) standing. Shooting rules are similar as in the relays. For first three (3) legs the change will happen right after the shooting and possible penalty loops have been finished. In the last leg there is one extra loop of skiing before finish line as in every other competition.

All the relay competitions are competed as mass start competitions and the winner is the first in the finish line. Mixed and single mixed relay are raced during the same day and single mixed relay was added so more athletes could compete in mixed relay day. Mixed relay was added to give smaller countries a change to compete in relays and later adding the single mixed relay gave this chance to countries which won't possible have even two athletes per gender competing in IBU World Cup circuit in regular basis. The investments of IBU for helping smaller countries in more ways than adding these competitions is seen especially in single mixed relays as in some weekends the list of starting countries are getting close to the max of 30 countries.

2.2 FIS Cross-Country

Over the years in FIS Cross-Country World Cup the schedule have been containing wider variety of competition formats than IBU World Cup. For couple years and also in season 2025-2026 FIS Cross-Country World Cup schedule has been more streamlined and cohesive. In general there are three (3) different distances for skiing races with mix of two (2) different skiing styles or sometimes having both of them in one competition. There also also team sprints, which are two athlete team competitions but there are no other relays on the season schedule. Due to the two (2) skiing styles there are fairly low amount of identical races for the year as all the three (3) distances are split into half [2].

In general there are usually slightly less athletes competing on each event compared to the biathlon. In women's competitions there are usually between 50-70 athletes with some cases having slightly more or slightly less. In men's races there are usually between 70-90 athletes. In total FIS cross-country World Cup contained 27 events, 16 distance events and 11 sprints. In distance events there was also bonus points awarded for the leaders of the race in certain parts of the track which counts towards overall standings [2].

The following chapters explain different competition formats citing the official FIS website.

2.2.1 Sprint

Sprint is maybe more of what the name suggests than a biathlon sprint. The sprint skiing is done usually in a track which is around 1,5 kilometers long. Depending on the track and conditions the goal time frame is between 2 minutes and 30 seconds and 3 minutes and 30 seconds. Sometimes, for example often in the Olympic Winter Games and World Championships, the track might be slightly longer and more challenging. Both women and men are racing the same track. This competition format is also ran with two different skiing styles: classic and skate.

In this competition format there are two phases: qualification and group stages. In qualification all the participating athletes will ski around the sprint track as fast as possible with 15 second starting intervals. Based on these times top 30 athletes will be qualified into the group stage. Based on the standings and the picking rules the athletes can pick their groups for the group stage.

The group stage is done in groups of six (6) athletes. From each heat two best will qualify into the next stage and 2 non-qualified athletes with best times will qualify as lucky losers. In group stage there are total of three (3) stages: first round (5 heats), semi-finals (2 heats) and final. Winner is the fastest to the finish line in final and positions outside the final are decided based on 1) the finishing position in the heat and 2) time.

2.2.2 Interval Start

In interval start competition format each athlete will start in 30 seconds intervals. In earlier season it varied how long the competition distance was but for now interval start competitions are always 10 kilometers long. The winner is who finishes the distance in fastest time. The competition format is ran with two different skiing styles: classic and skate.

2.2.3 Mass Start

A mass start format is very much similar to biathlon mass start format except there is not similar limitation for participating which means that everyone who are allowed to start in World Cup competitions can participate. The competition distance on this format is currently 20 kilometers. The winner is the fastest to the finish line. It's ran with three different styles: classic, skate and skiathlon. Skiathlon is a combination of the both styles. The competition starts with 10 kilometers with classic style, then all the athletes change their skis and the competition is finished with 10 kilometers with skating style.

2.2.4 Special events

During the year there's also Tour de Ski, a special combination of competitions which counts as a separate tour but also counts towards the World Cup but with modified points and prize money distribution. It's schedule varied year-by-year and it contains the regular distance competitions but also sometimes there are more special competitions in terms of distance and format. Tour de Ski is held every year around new year.

Other specialty is a Holmenkollen 50 kilometers competition which stands out from the rest of the schedule by it's distance. It has legendary status and when the World Cup schedule was revamped to it's current form, this competition maintained it's position in a calendar in it's original form. The only change for this is that also women are now racing the 50 kilometers as they used to ski 30 kilometers before.

Cross-country is also part of the Olympic Winter Games, which means that in season 2025-2026 it was the major event in the calendar. There is a differentiating thing compared to biathlon calendar however as the cross-country World Championships are held only once in two years. This means that every fourth year there isn't a "major" event in the calendar. Usually in these years there is more emphasis on competing in Tour de Ski.

3 Ranking systems and prize money distribution

3.1 IBU World Cup

In the IBU World Cup circuit there are running several different ranking systems parallel to each other. IBU also awards the best athletes/teams of the each competition with prize money. The summary of the explanation can be found here [7].

Total Score is the individual competition for the whole year which determines the best athlete of the IBU World Cup each year. It contains all the individual competitions, 21 in season 2025-2026. It's the most visible part of the competition circuit. Total Score point distribution saw a major reshape 4 season ago and has been tweaked last time for the current season 2025-2026. For each race top 40 athletes will collect points.

Nations Cup is the other ranking system which is meant for the countries. It's main purpose is to determine the starting spots for the next year for each country but the best countries are also awarded with prize money based on their performance. Nations Cup is not looked into more in-depth in this analysis.

On top of these each individual discipline is also followed separately with the same points distribution system as Total Score. They also have rankings for relays and mixed relays. Winner of all of these disciplines are awarded at the end of the season.

IBU also awards best teams with prize money for each race and there are some end of the season bonuses and other bonuses as well. In each individual race top 30 athletes will get awarded with prize money and in relays top 8 countries do as well. At the end of the season top 10 best athletes will get a bonus and winners of each discipline will also get a prize money bonus.

3.1.1 Ranking points distribution from season 2024/2025 onwards

The current Total Score and individual discipline points distribution has been used for 2 seasons, but a major overhaul was done for the season 2022/2023 and changes made for season 2024/2025 only slightly improved the points awarded for positions between 3th and 9th. Total of top 40 athletes will get points from each race. For the mass format the point distribution is modified from the bottom (from 21th to 30th) to evenly reduce the points towards the zero. In the new system winner will get 90 points and second 75 points, third gets 65 points. Currently the top 10 will get following points:

Position	Points
1	90
2	75
3	65
4	55
5	50
6	45
7	41
8	37
9	34
10	31

From this forward each position is worth of one points less until 40th will get one (1) point (expect in mass start). Positions from 10 forwards has remained the same for the whole change period so technically it's possible to compared seasons from athletes without any modifications for decades back if they have not finished any times in top 9.

3.1.2 Ranking points distribution from season 2022/2023 to 2023/2024

Before the minor tweak for the positions between 3rd to 9th for season 2024/2025 the distribution was slightly more top heavy. The minor tweak did give more value for positioning in most of the top 10 positions compared to the winner. The top 10 point distribution on these seasons were followed:

Position	Points
1	90
2	75
3	60
4	50
5	45
6	40
7	36
8	34
9	32
10	31

3.1.3 Ranking points distribution before season 2022/2023

Coming into 2022 IBU wanted to rewards the top positions with more points compared to the rest of the competitors to make winning more valuable in this fashion. Before this the distribution did have less separation between winner and 11th. Before the winner got double the points to 11th, but in new system he/she gained 3 times more points. Originally only top 6 positions gained a buff which then were tweaked later as we saw in the last chapter.

Before the major overhaul the top 10 distribution looked as followed:

Position	Points
1	60
2	54
3	48
4	43
5	40
6	38
7	36
8	34
9	32
10	31

It's good to note again that points distribution outside top 10 has been remained completely the same through all these changes. The other thing what is good to mention is that before 2020/2021 season there was also rule in place where two of the each competitors worst results were eliminated from the Total Score final results. This was originally done due the deviation what the sport brings especially with shooting element. This was often used by top athletes to rest and practice for one competition weekend during the season while preparing to Winter Olympic Games or IBU World Championships or many cases the athletes might have been more cautious of starting if they've had some ill symptoms.

With the major overhaul this rule was removed. IBU did want their star athletes to attend all competitions during the season, although Winter Olympic Games have still seemed to be such a major event in athletes calendar that even with this change most of the top athletes still skipped at least one weekend from the season.

3.1.4 Prize money distribution

IBU has also contributing for the athletes success by awarding them prize money in various ways. Total amount of prize money awarded out for the athletes in various cups is roughly 8,5 million euros. In IBU World Cup there are multiple performance based awards for each race and also so called bib-bonuses for starting as a leader of different competitions.

The main way to gather prize money with successful performances is to finish in top 30 in individual races or top 8 in any relays. In the season 2025-2025 winner of individual race is awarded with 20 000 euro prize and award declines down to 250 euros which is given to 30th finishing athlete. In relays winner team gets 30 000 euros and the last awarded place of 8th receives 2 400 euros. For single mixed relay this the relay prize money is halved due to team consisting only half of the members of a normal relay. In this way the individual of the team will receive always same prize money from same finish in any relay competition.

In the years when Olympic Winter Games aren't organized, IBU organizes IBU World Championships where the prize money from individual races are slightly higher than from normal IBU World Cup race. For the season 2025-2026 each IBU World Cup race was worth closer to IBU World Championships as the prize money reserved for that was shared in the IBU World cup. For example in previous years the individual competition winner has granted a 15 000 euros for their success and the rest of the distribution was adjusted accordingly.

On top of these there are some other bonuses. The main ones are the end of season Total Score result bonus which is awarded to top 10 athletes in the points. Winner of each smaller cup (sprint, pursuit, individual and mass) will also receive a bonus award. There is also bonus award for the best U23 athlete and nations will get a bonus from winning relay and mixed relay cups and performing well in Nations Cup.

But there's more. During the season there is yellow bib for the Total Score leader, red bib for the each discipline leader and blue bib for the best U23 athlete in Total Score. The athlete who happens to lead these competitions and is starting to the next competition is awarded a "bib-bonus" for each start they wear these bibs. Theoretically it's possible to wear all these bibs and then you get all the bonus start awards from that specific competition. Newer additions are also bonus awards for fastest pursuit time and fastest leg on relays and best woman/man in single mixed relay which combines the two legs each are competing in each race.

The whole 8,5 million euros doesn't go the top level. Secondary level IBU Cup and it's main event IBU Open European Championships are also getting their fair share of the prize moneys. This and the whole prize money distribution and amounts are covered more in-depth in IBU website [8].

3.2 FIS Cross-Country

As for IBU World Cup, in FIS cross-country World Cup has also several different ranking systems running parallel to each other and it too has prize money for the best athletes. The goals of these systems are more or less the same but they some some slight differences. As for this analysis the individual points distribution are used only for comparison on IBU results, so they will be introduced only briefly. Prize money distribution is covered slightly more in detail. The more detailed information for this can be found from here [3].

3.2.1 Ranking points systems and points distribution from season 2022/2023 onwards

The main competition is the overall cup which includes all the World Cup competitions and theirs results. The winner is the athlete who collects the most points during the year. From each individual competition total of 50 athletes will get points. FIS has made a change to the points distribution at the same time with IBU but they made the decision to go opposite direction. Win has always been worth of 100 points, but currently the places are worth 95, 90 points etc. as before the difference was much larger with 80, 60 points etc. The top 10 positions distribution looks as followed:

Position	Points
1	100
2	95
3	90
4	85
5	80
6	75
7	72
8	69
9	66
10	63

The decade from here slows down after 11th point to 2 points per position and from 31st position to 1 points per position resulting the 50th athlete go get awarded by 1 point.

There is also Nations Cup but as there are not as much competing countries and it's implementation to the tour isn't quite as the same at in IBU Nations Cup, it mostly flies under the radar.

3.2.2 Ranking points distribution before season 2022/2023

The old system awarded points for only top 30 athletes and the decay of points was much higher. The winner still got 100 points, but 2nd 80 points and third only 60 points. The top 10 looks as followed:

Position	Points
1	100
2	80
3	60
4	50
5	45
6	40
7	36
8	32
9	29
10	26

The decay continued as 2 points per position until the 15th position and rest of the distribution decayed by 1 points per position. The distribution is the most top heavy from all of the distributions used and introduced for this analysis.

3.2.3 Prize money distribution

As in IBU in biathlon, FIS in cross-country contributes to the best athletes with money prizes. In the season 2025-2026 around 4 million euros was awarded to the athletes, including the FIS cross-country World Cup and Tour de Ski. The there are multiple differences but one of the major ones, which requires mention is that there isn't a full tier-2 level international competition circuit. The next step underneath are few regional and national cups with less coverage over the year.

As for the biathlon, the each World Cup race has it's own prize money pool. Top 20 athletes in individual races and top 10 teams in team sprints will be awarded with prize money. In the season 2025-2026 the winner of the individual race will get 15 000 euro prize and the last prize awarded for the 20th place is 500 euros. In team sprints winner team will get 16 000 euros and 10th 1000 euros. All the other prizes are in between.

On top of this at the end of the season there is prize money awarded based on the overall standings. Also here top 20 athletes will be granted some prize money ranging from the winner, who receives 55 000 euros to 20th, who receives 1000 euros. In Tour de Skis independent events only top 3 will get a smaller award but for top 20 who finish the whole tour will get awarded starting from winner, who gets 80 000 euros. There's also couple minor bonuses for top athletes in best sprinters and best climbers.

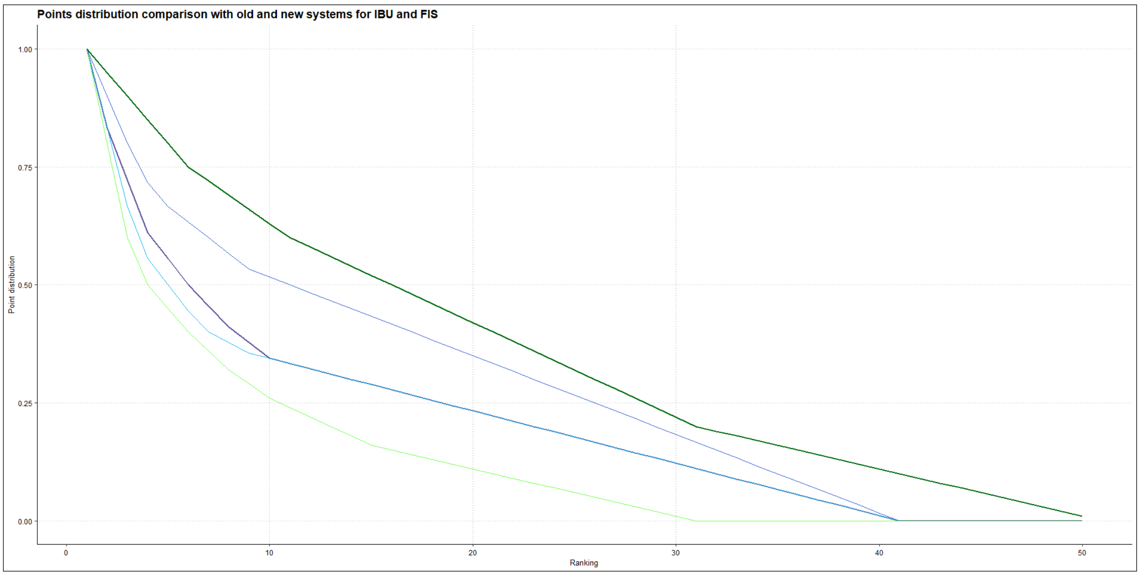


Figure 1: Comparison of all the different point distributions introduced for the analysis. Blue distributions have been used in IBU World Cup while green distributions have been used in FIS World Cup. The current distribution is in bold.

4 Pre-analysis comparisons

Before starting to do a major analysis it's good to do couple general comparisons between the sports. In this section the focus is on exploring the differences of each points distributions introduced in earlier chapters and comparing the prize money distributions of each sport.

4.1 Points distribution comparison

In previous chapter, total of 5 different distributions were introduced which have been used in either biathlon of cross-country skiing world cups over the last few decades. Before making the analysis of their effect on results of actual competitions, let's look how the distributions compare to each other in theoretical level. In figure 1 all the different distributions are showcased. The blue distributions have been used in IBU World Cup while the bold line is the current version in use. The green distributions have been used in FIS World Cup with bold line being current on their competitions.

The first and most notable observation is obviously how FIS overhaul of points distribution was very radical compared to the IBUs changes. As both changes happened during the same off season, it was really surprising to see that each organization went totally different directions with their changes. Overall the IBUs change was drastic but it effected more the top positions. The tweaked version of the distribution was also a correction towards the old distribution.

Meanwhile FIS overhaul was basically complete from top to bottom. The old distribution was the most top heavy of all distributions but it also did award ten less athletes so in that sense, that could partially explain this top heaviness. After the overhaul the new distribution was least top heavy and it also awarded points for twenty athletes more who didn't get any points before. The changes of results have not been compared in this analysis, but it must be said that in the field of competition FIS World Cup presents, this change have had a major impact on the end of the season results.

In both cases the changes weren't necessarily taken by open arms by the communities, although in IBU World Cup the bigger discussion was the removal of another minor rule which effected to the total standings at the end of year. After the major changes IBU has taken it's chance to tweak their system back closer to their older distribution while FIS has used the same system since the major change up to the end of season 2025/26.

4.2 Prize money distribution comparison

Before analysis part of the report it's good to highlight some key factors and compare these two distributions with each other.

First, it's important to mention that in IBU World Cup the Olympic year was a special year. In usual year for the recent years, the winner has been awarded with exactly the similar amount than what FIS cross-country World Cup winner receives in this year. As in a usual year IBU will also award prize money in the World Championships the individual competitions receiving the prize money is roughly the same, unlike this year. FIS doesn't award any prize money in their World Championships. It's still good to point out that in a usual year in IBU events on top of the individual competitions there are roughly 10 relays which athlete can participate to gain prize money when in FIS events there are only couple.

The main difference between the sports is that in biathlon there is over double the prize money which is awarded over the year. Partly due to this reason it wouldn't be useful to compare these with each other cause in most cases throughout the whole athlete pool the biathlete will receive more prize money compared to the same ranked skiing athlete. This is also why it's better to compare the prize money distribution in percentages of the whole prize money awarded during the year and look if there are differences in the spread out between these sports. To achieve same amount of money, the athlete in the similar position in the ranking should receive almost twice the money from whole pool in FIS World Cup than in IBU World Cup.

From technical point of view there are also major differences. In biathlon in every individual race 10 more athletes will be awarded which in theory should distribute the money for more wider group of athletes. In cross-country in team events more teams are getting prize money, although in the end it doesn't cover as many athletes than relay days in biathlon (20 athletes in team sprint per gender in cross-country vs 24 athletes in mixed relay days or 32 athletes in relay days per gender in biathlon). The bigger prize money pool gives also opportunity for biathlon to share bigger prize moneys for lower finishing athletes in the race. In season 2025-2026 if you placed on 6th place in the individual competition (in 2024-2025 5th place), you got awarded more prize money than if you finished 3rd in cross-country. This continues mostly throughout the whole distribution but the biggest dropout in the FIS World Cup is definitely after the winner and 2nd place. The fact that less athletes are awarded prize money per race and the drop is way more drastic it should mean that less athletes should get prize money.

The other differences comes at the end of the season bonuses. In FIS cross-country World Cup winner is awarded more prize money but it also awards much wider range of athletes, although with much smaller prizes. The single most valuable event to win is still Tour de Ski, which winner receives 80 000 euros worth of prize money. This distribution is also fairly top heavy but it's still significant as 10th is still receiving 9 000 euros. It's good to mention a certain issue of the distribution which is that individual Tour de Ski events will only grant prize money to top 3 athletes. As there was six (6) competitions part of the tour this year, finishing 10th granted less money than finishing six (6) times as 10th in normal FIS World Cup competition. Also IBU awards the discipline winners with prize money at the end of the season, which FIS doesn't do.

5 Analysis

The goal of the analysis is use visualization as a tool to understand and make observations from the results. Analysis will cover overall result analysis of IBU World Cup and tries to answer to the questions of how the current points distribution shaped the results, and what types of participation it would favor, if any. Analysis also includes point distribution comparison based on last 5 seasons of IBU World Cup results including all five distributions discussed in previous chapter and prize money distribution comparison between IBU World Cup and FIS World Cup.

The information is gathered from IBU Datacenter and official FIS website. Some of the features are available straight from the source and in the case of FIS related data it has been used without further modifying. with data from IBU Datacenter the straight collection of data didn't give the opportunity to do much so all the data from the biathlon races are gathered individually from the same source, modified and combined for the need.

5.1 IBU Women's Total Score results

The main discipline in the IBU World Cup is the individual overall competition which includes all the indivial races from the year. As mentioned before, it contains 21 races in season 2025-2026. From the official IBU website we can gather that the winner of this years women's Overall competition was Lou Jeanmonnot from France [9]. She gathered total of 1135 points and had fairly big margin to the others. During the season total of 172 athletes were competing at least in one race and 102 out of them managed to finish at least once in top 40 to gather some points.

First look will be the distribution of the points from top 30 athletes from the season 2025-2026 divided into two graphs and at the end we take a look into development of top 10 athletes and how they ended up into their final results. All the specific information about competition results can be found through here [9]. If there is a mention about a specific athletes compettiton results, you can find them by clicking their name to open their profile and look for latest results. Some statistics mentioned are only available through the data

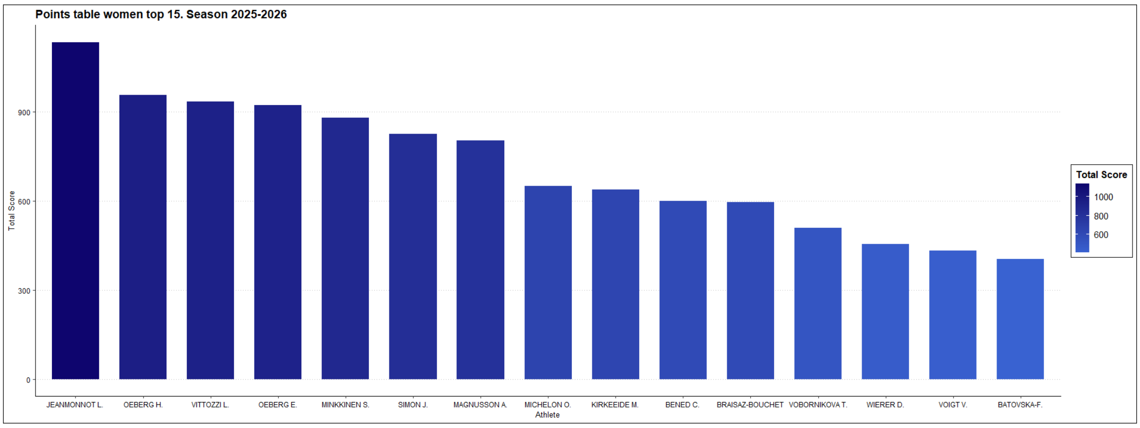


Figure 2: Points from women's top 15

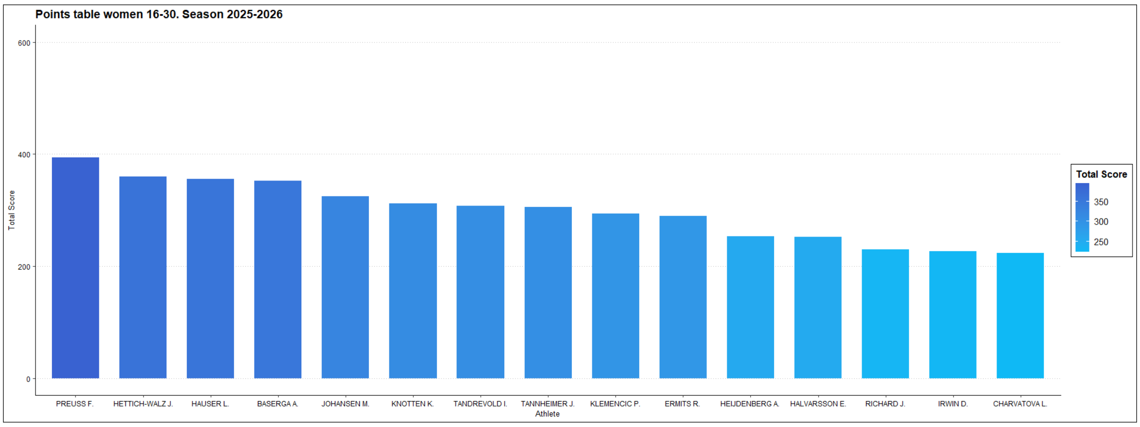


Figure 3: Points from women's top 16-30

processing.

Figure (2) shows top 15 athletes of the season. The figure shows fairly big gap between the first and the rest but there is another gap after top 7 athletes suggesting that these athletes were the ones competing from the highest position with the highest consistency. Based on points distribution on single events and compared that to the total points, the rest of the athletes had fairly good chance on fighting from top 3 positions but they couldn't do it consistently. Either their consistency or overall level was one step behind top 7 athletes.

Figure (3) shows next 15 athletes. Just as expected the gaps between the athletes starts to become smaller. Most of these athletes did have chance to surprise in single race to gather bigger points but they didn't have the same consistency for fighting from the absolute top positions throughout the season than the group above on positions 8-15. Greatest example of a big surprise from this group would be Lisa Hauser winning a Östersund Pursuit after finishing 4th in the preceding Sprint competition. Including both races, she didn't miss a single target, which she wasn't able to reproduce in the rest of the season.

To make analysis more in-depth some extra features should be added to understand the results better. To do this for the next two figures the gradient of the bars shows average position of the athlete throughout the season. For more information a line graph with dots

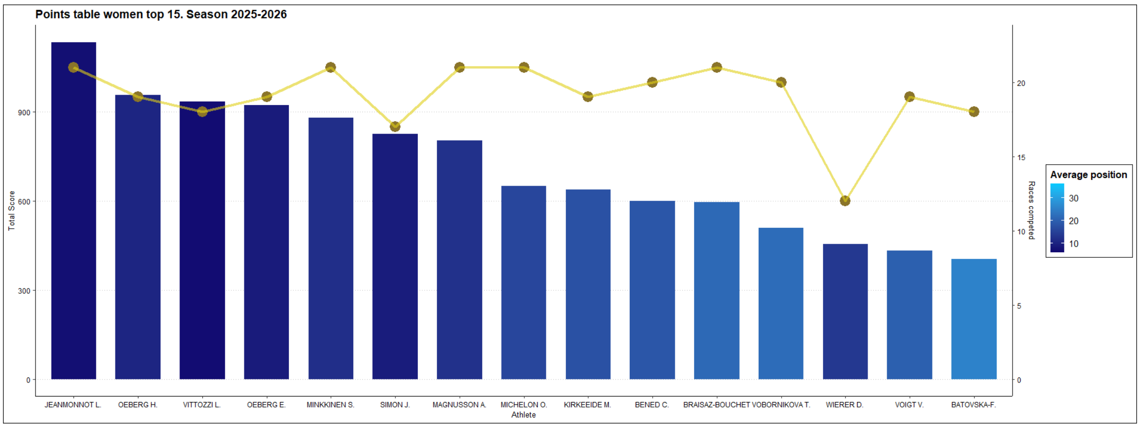


Figure 4: Points from women's top 15 with added gradient based on average position and dots and line showing the amount of races competed.

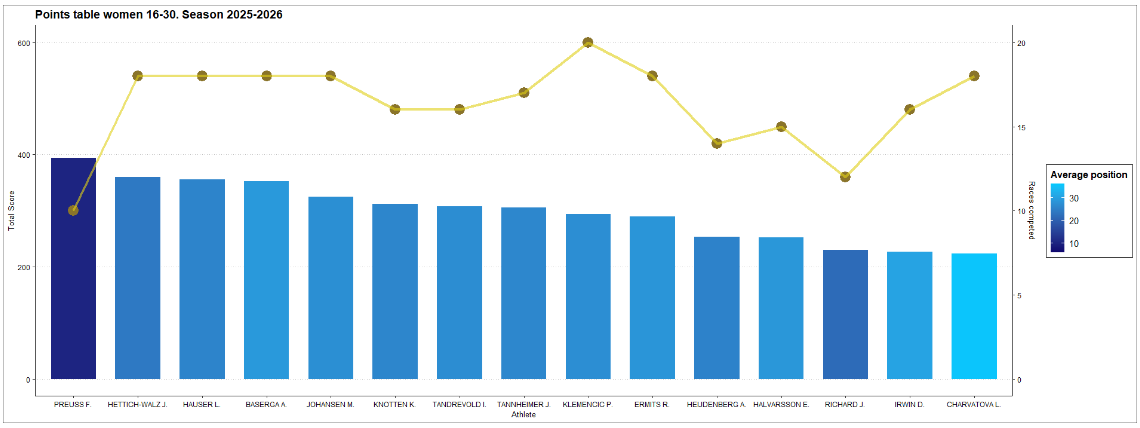


Figure 5: Points from women's top 16-30 with added gradient based on average position and dots and line showing the amount of races competed.

was added to showcase how many races each athlete participated in the season 2025-2026. Y-axis of the races competed can be seen on the right side of the plot. To make analysis possible in between of the figures the range of the gradient scale has been adjusted to be the same. The darker color signifies the higher average position.

The changes in the gradient in the figure (4) are fairly subtle but there are some things to gather. When combining all the three information, total points, average position and races competed, it's possible to make deeper observations. Clear outlier in the top 15 is Dorothea Wierer who finished her career after Winter Olympics and didn't compete in the last trimester. Due to this fact she didn't participate as many races than others in average but she had a very good season with clearly higher average position than athletes around her. Other than this outlier the top 7 separates from the rest of the group while having subtle differences. In general it's possible to say that if athlete competed more often, she had slightly weaker average position compared to the athletes finishing close to similar points. Also it's good to mention the winner Lou Jeanmonnot reaching a average position of 7.2 which also wasn't quite the highest as Liza Vittozzi manage to top that statistic with 6.8.

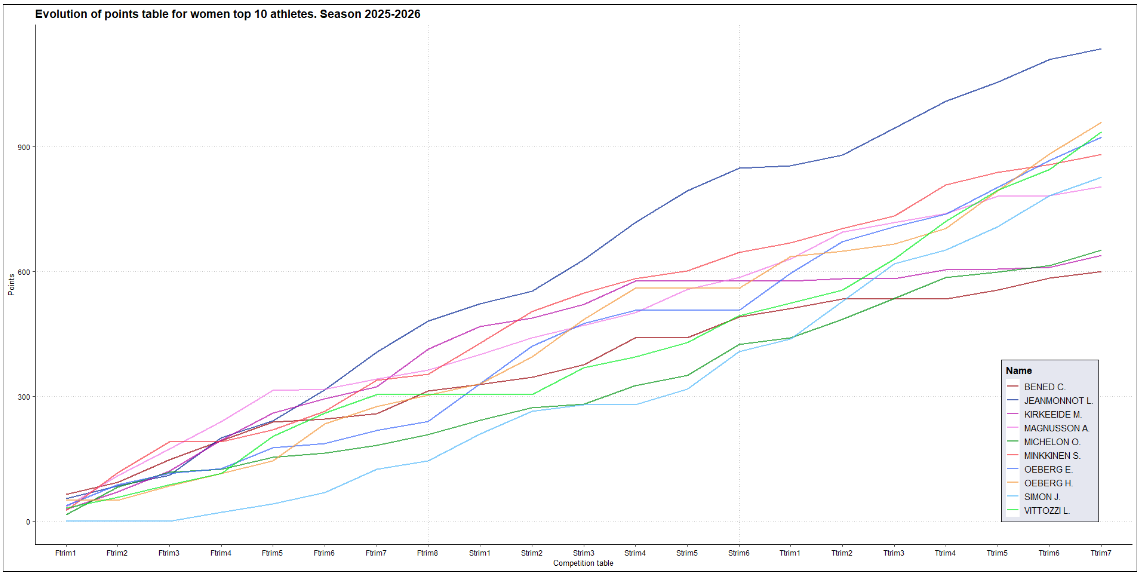


Figure 6: Development of the points table from the top 10 women athletes at the end of the season 2025-2026.

From the second figure (5) it's possible to find another clear outlier but possibly also another smaller outlier. Franziska Preuss had similar decision to Dorothea Wierer and she also finished her career in the Winter Olympic Games. On average Preuss was performing even higher standard as her companion, but on top of the missing third trimester she also missed some competitions from the start of the season. In total she did compete in less than half of the all individual competitions during the season. If competing for the whole season, both Wierer and Pruess were clearly capable of competing in the level of top 7 athletes, however Wierer had more inconsistency on her results. Based on this analysis they can be added into the same group with top 7 athletes to create the top 9 athletes who separated from others, when they were on the start line.

Before other outlier the colors are signifying what were expected already based on points. Most of these athletes 5 weren't able to compete consistently from the top positions, which shows as higher average position per race. These athletes also had few less competitions on their belt during the season compared to top 15. The reasons may vary from health issues, preparation to Olympics, missing mass starts or in some cases there could be another issue which French athlete Jeanne Richard was facing. She is the another outlier in top 16-30 and her problem was to fit into French team due the restrictions of 6 athletes for their country. The competition from these spots was very high as France is currently having the most amount of high level athletes and they all can't fit into the World Cup competitions at the same time. She was performing better than athletes around her but she was forced to give her spot in Annecy and the whole last trimester for other athletes. If competing the whole year she could've been part of the top 20 athletes or even higher.

Looking into the development of the season from top 10 athletes 6 it's possible to see couple different progression styles. There were couple early season successors, including Anna Magnusson and Suvi Minkkinen, who claimed the top position in Total Score mostly performing well in the first part of the season. Lou Jeanmonnot managed to overtake them already still during the first trimester and claimed majority of her lead in the second trimester compared to the others. Hanna Öberg did have a good result on very first competition but had slightly slower mid season until the end of third trimester where she managed to overtake 4 athletes in last 3 competitions to claim second place in the standings. Julia Simon was the late entry for the season starting only the second weekend but third trimester was the best for her. Missing the first weeks action did drop her slightly behind from second to fifth place. Oceane Michelon had a stable year but her results were slightly weaker than the most until the end where she managed to overtake two athletes.

Visualizing the season like in graph 6 can give even person not following a peak to how season progressed. After first trimester it looked like athletes like Maren Kirkeeide, Magnusson and Minkkinen, who were new names for the fight from overall win would be competing the hardest from it against the possibly biggest favourite Jeanmonnot. Meanwhile many others, who are considered as star names, ended up pushing to top 6 at the end of the season but didn't really compete from the total victory at any point. The third trimester did show that with having a great success during one of the three trimester can still push you near to the top but if your season is quite not consistent on high level, winning the overall is nearly impossible. It also showed that a lot of the biggest stars did prioritize the Winter Olympic Games over balanced performance in the IBU World Cup and this is also maybe a reason why Jeanmonnot was able to win the overall with comfort while maybe not performing not as high level than year before.

5.2 IBU Men's Total Score results

As for women, also men had 21 individual races for the season 2025-2026. Men's winner is Eric Perrot and comes also from France [9]. His gap for the rest of the athletes was even bigger than Jeanmonnot gathered in the women's side. In men's race there was some more unfortunate events during the season other than skipping some IBU World Cup weekends to prepare for the Olympic Winter Games or athletes retiring before the season ending. A surprise death of an athlete during Christmas break and serious heart problems effected on handful of top athletes to skip some of the competition and most likely the effected the Total Score race in major ways. Credit has to be given to the winner Perrot even so, as he was super consistent in the top level over the whole season, which will be covered in a moment. Again, the personal competition results can be found from [9] but some statistics are only available through the analysis.

In the season 2025-2026 total of 187 men competed at least once and total of 105 of them manage to finish one or more times in top 40 to gather points for their Total Score results.

In the first figure (7) the top 15 athletes on men are introduced. In men's competition it's not quite as obvious to notice a clear gap in points to create groups from the athletes inside the top 15. Interpretation could be to separate top 5 athletes from each other but Italian Tommasso Giacomel is sort of in between of the two groups. It's clear that more information is needed for better analysis.

Looking into second figure (8) and this information to what was in the first figure (7) the clear gap between athletes is now visible. After Fabien Claude in 16th place there is clear

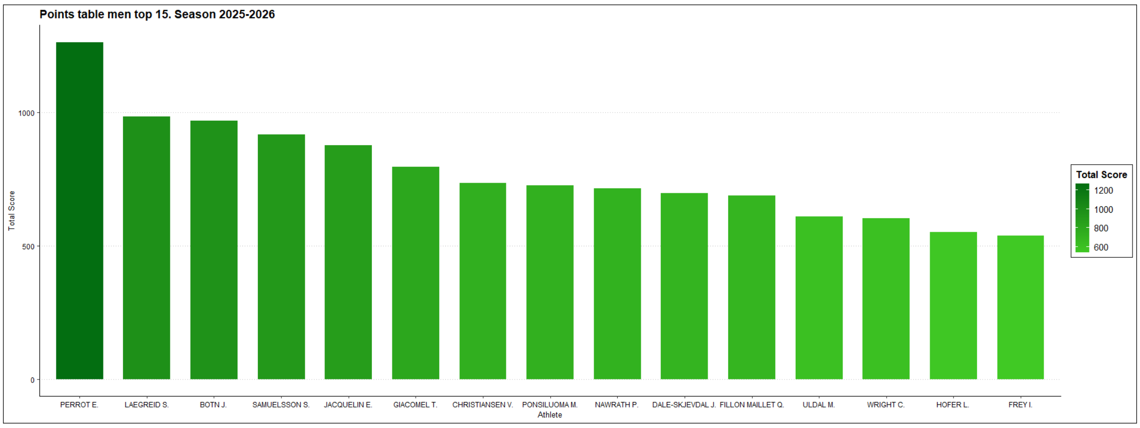


Figure 7: Points from men's top 15

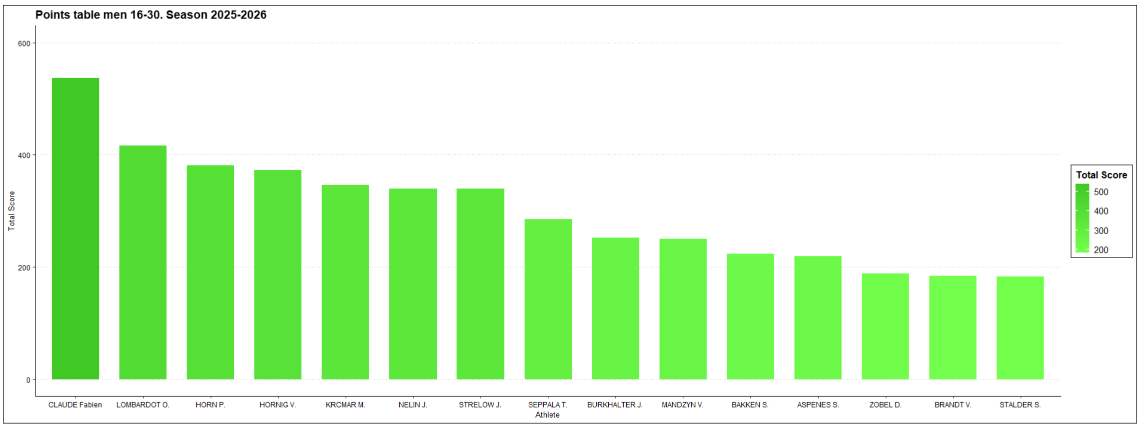


Figure 8: Points from men's top 16-30

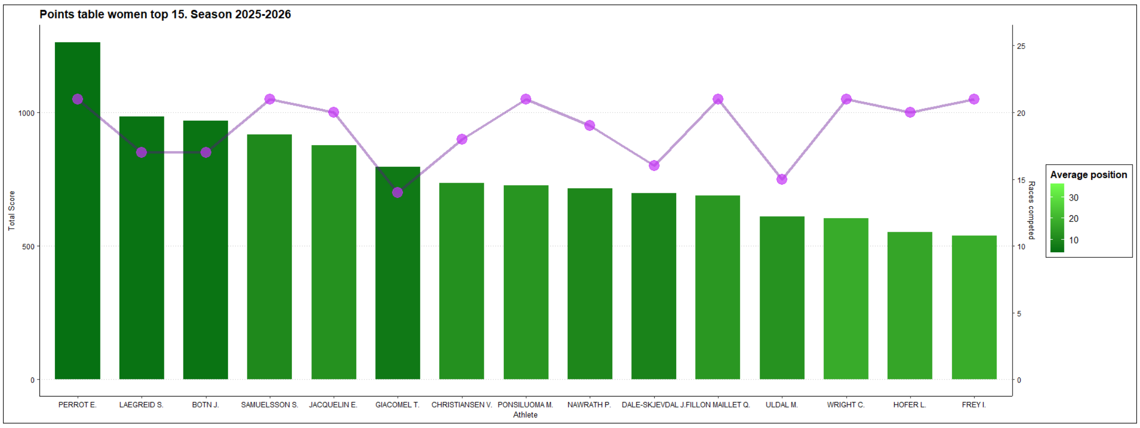


Figure 9: Points from men's top 15 with added gradient based on average position and dots and line showing the amount of races competed.

gap between points. The rest of the athletes are more of less in the pack although it could be argued that since the point distribution is smaller after top 10 on each race that after Claude there's another group of 6 athletes and a gap for the rest of the athletes as well. Based on knowledge outside of pure data crunch, this gap basically is created as the top 22 athletes were able to gather at least few top 10 positions which were more valuable with point distribution. There are of course few exceptions to this rule which will be mentioned after improving the visualization.

To improve analysis the gradient is changed to the average position and the race total is added to the figure. The gradient is again the same over the two following figures.

(9) With updated gradient the figure is not the easiest to analyze as the top athletes perform clearly better than the athletes in the graph (10). With closer look it's possible to separate four athletes who were performing better than the rest in terms of average position. These athletes are Perrot, Sturla Holm Lägreid, Johan-Olav Botn and Giacomel. Giacomel out of these four had to skip the last trimester due to the health issues. The main reason why the gaps between these athletes are so big is the amount of competitions they were participating throughout the year.

From the rest of the top 15 it's possible to recognize two more athletes who performed slightly better than their closest rivals and they are Johannes Dale-Skjevdal and Martin Uldal. They also had less competitions on their belts over the season and the reason for this based on further analysis is that Dale-Skjevdal skipped a weekend to prepare for the Olympic Winter Games and Uldal most likely had some issues with his health, leaving him out from one of the weekends at the start of the season. If competing the whole season they probably would've been slightly higher in the final results but still behind the top 4 athletes.

When looking for the rest of the top 30 in (10) the gradient predictably becomes lot brighter. The most notable athlete is Sivert Guttorm Bakken who competed only two weekends and had clearly better success compared to the athletes closest to his points. By side of him there's also another Norwegian Sverre Dahlen Aspenes who also competed

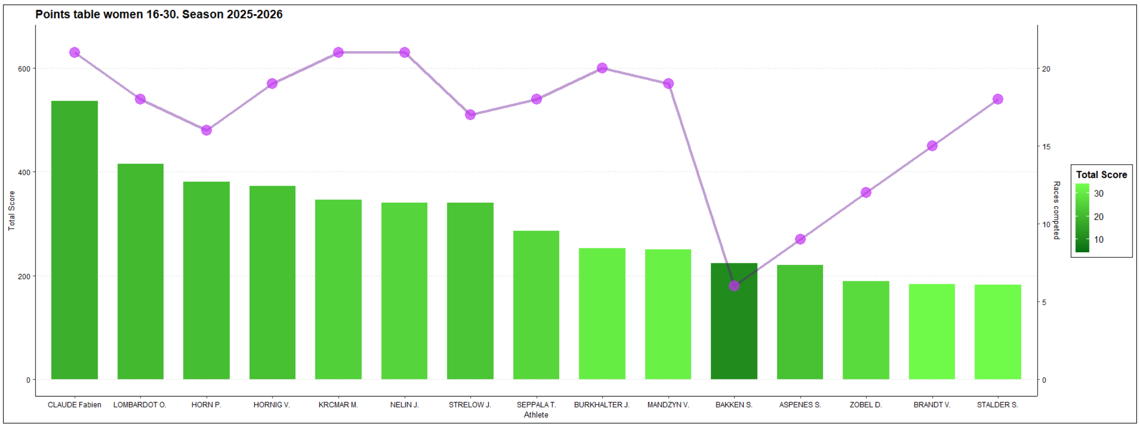


Figure 10: Points from men's top 16-30 with added gradient based on average position and dots and line showing the amount of races competed.

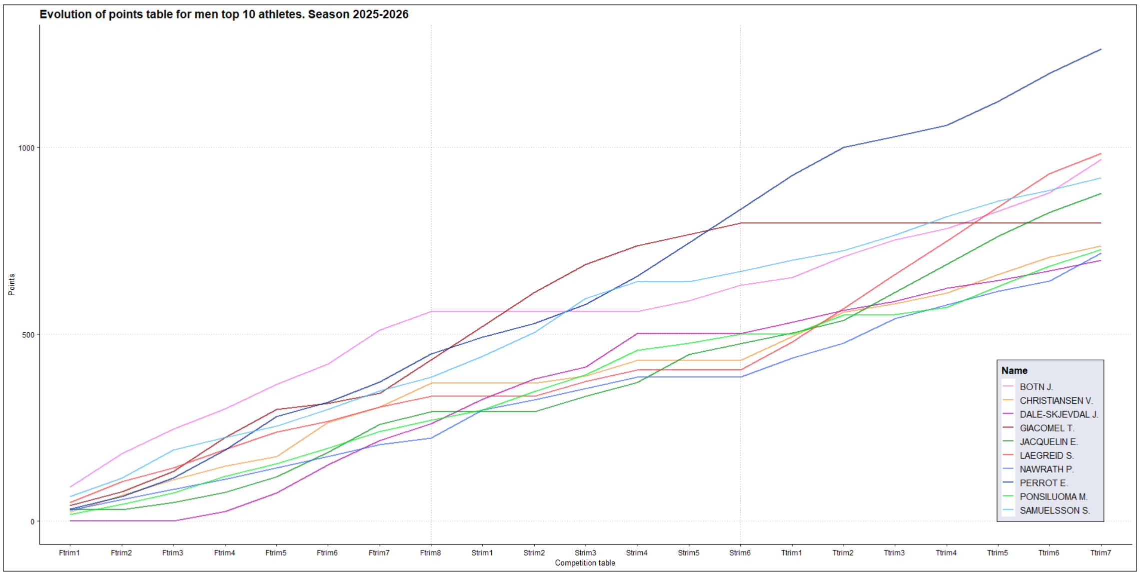


Figure 11: Development of the points table from the top 10 men athletes at the end of the season 2025-2026.

only few times during the season and he was facing similar issues than Richard in women's side, national competition. He did also have fairly poor performances and almost half of his points came from two competitions in Nove Mesto.

Looking at the development of the competition of overall in men's top 10 11 it's possible to see how Botn was basically dominating the first trimester, Giacomel took the lead due Botn missing some competitions and performing slightly better in second trimester than Perrot but eventually it was Perrot who took the lead for the end of the season as Giacomel wasn't able to compete. Also Sebastian Samuelsson was close on the race to overall win mid-way to the season but wasn't able to keep the level of Perrot at the second half. The rest of the top 10 did perform fairly evenly until the third semester where Lägreid was able to peak from the rest and finished to the second place in the overall by winning all but two competitions after the Winter Olympic Games and finishing worst at fourth. The

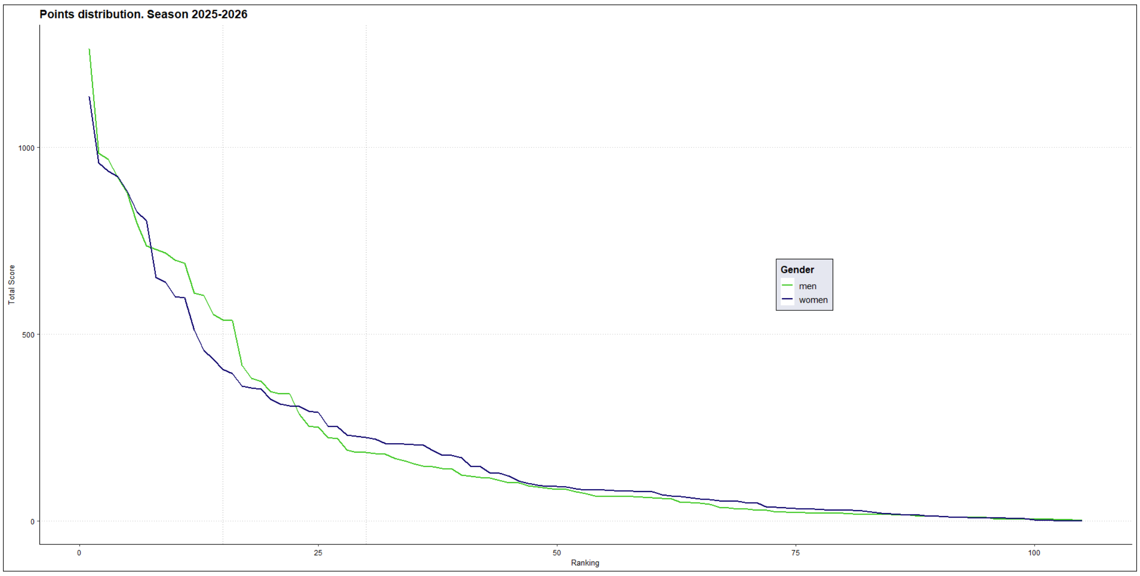


Figure 12: Comparison of point distribution on season 2025-2026.

main reason why he wasn't able to catch the winner Perrot more was because of Perrot's great consistency.

In general both women's and men's figures showcases that usually top 15 athletes had on average more competitions but the margin isn't that high that it would be significant. Apart from the preparation for the Olympic Winter Games, health issues or not making the team being the reasons to not start, this can also be due to the fact that only top 25 athletes are allowed to start in mass competition and some of the athletes at the bottom of the analysis were not eligible to automatically participate in all the races. It's also more likely that athletes around top 30 in the overall standings had slightly higher chance of missing the pursuit competition after poor performance in sprint compared to the top athletes. In men's side the group of athletes who were able to compete consistently from top 6 finishes where slightly wider which also created a major gap after this group to the athletes who barely were able to reach into top 10 during the season to gain bigger chuck of points from one finish.

5.3 IBU Points distribution comparison

As the season structure and point distributions are equal between genders, it's possible also to compare the results in between of each other. This analysis can reveal differences in between two competitions and it's also doable over seasons. For now the comparison is made from season 2025-2026 overall results.

As mentioned earlier, 102 women and 105 men were able to collect points during the season. Total amount of competitors was also relatively close, 172 for women versus 187 for men. For the next graph all the athletes who managed to collect points were included. For additional help two vertical dotted lines were added on 15th and 30th place in comparison to make visibly more clear the comparison earlier made gender specifically.

First observation looking at graph (12) is the obvious one where winner of the men's overall competition scored more points than women's winner. This was mainly possible because of higher competitive consistency. Other than this fairly clear difference the top athletes were very comparable between each other in terms of points gathered. First major difference in the distribution comes when women's distribution has the first major gap discussed about earlier after 7 athletes. From this point men's athletes systematically were able to collect more points on average up to around 20th athlete. This can be seen as a sign from wider competition in men's side right after the top athletes. This was also possible to see in men's figures (9) and (10) where the first clear gap was after 16 athletes. Even this drop wasn't yet enough to make women's athletes to collect more points in comparison but instead the next gap discussed before finally made it.

The second major cut off point is after 20 or so athletes when men's distribution has bigger gap. After this change women's athlete consistently collected more points until the very end of the graph where the last few athletes are represented who manage to collect just few points, although this gap closes in already earlier. This would still suggest that in women's side the mid level competition is more equal than in men's side.

One reason could be also that in men's competitions there are more athletes who are capable for top results. Finishing one time in the podium per season while performing in solid level for the rest of the season, usually results over certain threshold of points, which is difficult to achieve if athlete is consistent in top 20, but can't break into top 6 or podium. This leads into situation where there is less to gain for the rest.

In women's side the competition is maybe not quite as wide in the absolute top which then gives more opportunity for wider group of athletes in the mid range of the pack. In technical terms it still applies that even one or two top performances with solid consistency will pull an athlete over certain threshold but in women's competition there can be more standard deviation for certain athletes. One example of this is Finnish athlete Sonja Leinamo who started the season with excellent second place finish in the first individual competition but after this she couldn't finish to the points more than once in other individual competitions. She for example couldn't finish into top 60 in any of the sprints, a hard line she had not cut more than once on season before either. Because of this she is an outlier of one podium performance but not even close of able to be part of out earlier analysis of top 30 athletes of the year.

In general it can be said that there are maybe more of a mix in the women's competitions right after to absolute top and there are also more women who are capable at reaching even top 15. When in men's competition there are more athletes who are able to peak at higher level, it does leave less room for others even when they would do near to perfect competition while others with peak capability would not. One metric that was measured but not very visible in the analysis, was standard deviations of the positions over the year and average of this variable was slightly higher with women athletes than with men athletes. This could in theory suggest the claim but the difference wasn't very high (women 15.4 vs men 14.4).

5.4 Comparing points distributions effect on top positions

One interesting comparison which can be made is to compare different points distributions with each other and see if they end up shaping the end of the season results differently when the individual competition results are the same. For this comparison all the three major IBU points distribution system from last decades are used and comparison is made with top 15 athletes on both genders. To make analysis even more depth the current distribution is compared also into FIS World Cup distributions, the current and the old one.

5.4.1 Women's top 15 with IBU related distributions from season 25-26

Lets first look at women's table of top 15 athletes. Note that for this comparison the rule which deleted the two worst results were not used to Pre 2020/2021 points distribution.

IBU Current	Points	From 22/23	Points	Pre 22/23	Points
Jeanmonnot	1135	Jeanmonnot	1095	Jeanmonnot	874
Öberg H.	958	Öberg H.	938	Öberg H.	738
Vittozzi	935	Vittozzi	908	Vittozzi	738
Öberg E.	922	Öberg E.	886	Öberg E.	727
Minkkinen	881	Minkkinen	862	Minkkinen	721
Simon	827	Simon	797	Magnusson	676
Magnusson	803	Magnusson	763	Simon	651
Michelon	651	Michelon	618	Michelon	571
Kirkeeide	639	Kirkeeide	610	Kirkeeide	528
Bened	600	Braisaz-Bouc	567	Bened	518
Braisaz-Bouc	597	Bened	565	Braisaz-Bouc	509
Vobornikova	510	Vobornikova	490	Vobornikova	466
Wierer	456	Wierer	434	Voigt	413
Voigt	433	Voigt	415	Wierer	373
Batovska-Fial	405	Batovska-Fial	382	Batovska-Fial	360

There are not too many differences on top 15 regardless of what points distribution system would've been used. Top 5 remained in same order although with original distribution 2nd and 3rd athlete did tie. Hanna Öberg had more wins so she would've stayed in the 2nd place. The differences between athletes did have quite big chances. The new system made Lou Jeanmonnot even more dominant, Hanna Öberg did gain over 20 point lead against Vittozzi, when old system did have them in tie, Elvira Öberg was over 40 points in front of Minkkinen (compared to old system with 6 points) and so on.

The only effect on top 15 which did take place due to the minor tweaks of 3rd to 9th place was that Justine Braisaz-Bouchet did overtake Camille Bened with tight margin. This was tight race in all three versions so it shouldn't be a surprise that this could happen. Based on average position and standard deviation Camille Bened did perform better so the technically points distribution should show this as long as they have equal amount of races for the season as well, which they almost had. Other major beneficiaries of the minor tweaks in top 15 were Elvira Öberg and Oceane Michelon who both gained a fairly big advantage against the next competitor in standings.

Comparing the current system with the pre-major overhaul there was two changes where Anna Magnusson overtook Julia Simon and Vanessa Voigt overtook Dorothea Wierer. These changes, compared to couple other notable differences, a clear sign of differences between the new points distribution and old points distribution. Also athletes like Vitozzi, Minkkinen, Michelon and even Batovska-Fialkova would've gained advantage from the old system at least somewhat even they didn't gain any places in the standings, but at least they got a lot closer to achieve those improvements. The old system clearly benefited those athletes who weren't quite at the top but were able to consistently perform on top 10 level and even top 20 level as it didn't have as massive loss of points if your hardest competitor manage to finish top 3 or higher. The new system benefited quite clearly those who were able to perform in absolute top level at least at some point of the season. Simon did have higher top results even she didn't race at the start of the season. Wierer managed to keep Voigt behind her even she didn't race in the last trimester at all. Voigt was fairly consistent at finishing in top 20 but wasn't really in contention to podium at almost any race which Wierer was able to do during the parts of season she was still competing.

5.4.2 Women's top 15 with FIS related distributions compared to the current IBU distribution from season 25-26

Next let's take a look into how the distributions used by FIS would've shaped the top 15 athletes standings. For the most part it was possible to just cut and copy the distribution into the IBU World Cup competitions but there was one exception which needed some modification. The Mass start at IBU has already some adjustments since the normal distribution awards 40 athletes but in the mass start there's only 30 athletes competing. The solution was to reduce the points from 21st onward by two to reach 2 points for 30th and imaginary 31st would then get 0 points. The old system used by FIS didn't need any modifications as it awarded top 30 athletes on every competition but with the new system the same philosophy was used than what IBU uses for their distribution on mass starts. That meant that points were reduced by 4 from 21st onward until the 30th got 4 points and imaginary 31th would get 0 points.

The top 15 athletes standings comparison with FIS points distributions are followed:

IBU Current	Points	FIS Current	Points	FIS Old	Points
Jeanmonnot	1135	Jeanmonnot	1625	Jeanmonnot	1098
Öberg H.	958	Vittozzi	1380	Öberg H.	942
Vittozzi	935	Öberg E.	1371	Vittozzi	894
Öberg E.	922	Minkkinen	1361	Öberg E.	861
Minkkinen	881	Öberg H.	1354	Minkkinen	814
Simon	827	Magnusson	1324	Simon	773
Magnusson	803	Simon	1223	Magnusson	694
Michelon	651	Michelon	1130	Kirkeeide	571
Kirkeeide	639	Kirkeeide	1036	Michelon	508
Bened	600	Bened	1027	Braisaz-Bouc	494
Braisaz-Bouc	597	Braisaz-Bouc	999	Bened	470
Vobornikova	510	Vobornikova	930	Wierer	387
Wierer	456	Voigt	842	Vobornikova	380
Voigt	433	Batovska-Fial	730	Preuss	330
Batovska-Fial	405	Wierer	715	Batovska-Fial	307

As shown in the table above, the points distributions used by FIS would've changed the standings quite radically. In the old distribution the top 7 would've still stayed the same but with the new distribution the places even in the top 7 would have been re-shuffled.

The major changes from the new system basically comes from three athletes going down in the standings and total of 6 athletes gaining from that drop off. The athletes who dropped are Hanna Öberg, Julia Simon and Dorothea Wierer. With latter two the drop can be explained with missing the competitions compared to their counter parts but with H. Öberg the reason is slightly most complex. She did attend to similar amount of competitions than Elvira Öberg and Lisa Vittozzi but the main issue against these competitors were that she was more inconsistent. Also Suvi Minkkinen overtook her in the standings with the current FIS distribution but this was due to Minkkinen not missing any races while having similar enough performance throughout the year. The race between the 2nd to even 6th was much more closer with this system as even Anna Magnusson was only 30 points behind the group mentioned above. She was also having a similar season than Minkkinen, attending all the competitions but having slightly poorer results with similar deviation.

The of points distribution used by FIS didn't make any changes to top 7 but after that there were few who would've benefited from this system. Maren Kirkeeide, Justine Braisaz-Bouchet and Dorothea Wierer all gained positions inside top 15 over Oceane Michelon, Camille Bened and Teresa Vobornikova. On top of that Franziska Pruess did jump into top 15 from 16th position and dropped Vanessa Voigt out from top 15. What was all common to these athletes were that they had better performance at some part of the year compared to the athletes who they surpassed and they also in general they usually had less competitions throughout the season. The biggest showcases from this were Wierer and Preuss.

5.4.3 Closure of Women's top 15 comparison

To close out the comparison of the different points distributions in context of one season results. The next table gives us the possibility for more open ended analysis which ties all the discussion above. Let's take a look into one more table, which shows the top 15 athletes, the numbers of races they competed, their overall position, average position in the results with the alternative points distributions, their average position per race and standard deviation of the position per race.

Name	Races	Ovr Pos	Alt Pos Avg	Avg Race Pos	Race Pos SD
Jeanmonnot	21	1	1	7.2	8.1
Öberg H.	19	2	2.75	10.4	13.8
Vittozzi	18	3	2.75	6.8	5.1
Öberg E.	19	4	3.75	8.9	8.7
Minkkinen	21	5	4.75	11.8	11.1
Simon	17	6	6.5	9.0	8.0
Magnusson	21	7	6.5	12.3	10.6
Michelon	21	8	8.25	15.8	10.7
Kirkeeide	19	9	8.75	17.6	17.4
Bened	20	10	10.5	18.4	15.7
Braisaz-Bouc	21	11	10.5	21.4	20.0
Vobornikova	20	12	12.25	21.9	19.3
Wierer	12	13	13.5	13.4	10.2
Voigt	19	14	14	19.9	11.6
Batovska-Fial	18	15	14.75	25.4	20.1

Here we can see some of the information, which has been mentioned already before such as Lisa Vittozzi being the highest performer by average position and standard deviation but due to lack of participating to the races, she ended up third in the rankings. This information isn't not enough to do full conclusion, which can be seen of numbers of Hanna Öberg, who had roughly similar average and standard deviation with Suvi Minkkinen and even less races she ended up few spots higher in the rankings, beating even Lisa Vittozzi with much better numbers and with only one more race competed during the season.

The interesting new value is the average of the rankings from the other distributions. As we saw earlier, in this case only distributions used in FIS competitions were making major changes to end results. There are three ties between athletes with averages and those showcases the most common switch ups in the ranking table. Other than these the averages are in line with the overall position, which is also expected since the distributions calculated in the average are around the current points distribution system. To gain more information, let's look the same numbers with men's competitions.

5.4.4 Men's top 15 with IBU related distributions from season 25-26

After looking the examples from women's side, lets turn to men's examples. The comparison style is equal. The table looks as followed:

IBU Current	Points	From 22/23	Points	Pre 22/23	Points
Perrot	1263	Perrot	1225	Perrot	957
Lägreid	984	Lägreid	948	Samuelsson	760
Botn	968	Botn	918	Lägreid	751
Samuelsson	918	Samuelsson	861	Botn	739
Jacquelin	876	Jacquelin	844	Jacquelin	710
Giacomel	797	Giacomel	779	Nawrath	636
Christiansen	736	Nawrath	695	Ponsiluoma	631
Ponsiluoma	727	Christiansen	688	Christiansen	613
Nawrath	716	Ponsiluoma	675	Fillon Maillet	613
Dale-Skjevdal	697	Dale-Skjevdal	670	Giacomel	605
Fillon Maillet	689	Fillon Maillet	669	Dale-Skjevdal	581
Uldal	609	Uldal	589	Wright	551
Wright	604	Wright	581	Uldal	505
Hofer	552	Hofer	529	Hofer	505
Frey	538	Frey	517	Claude F.	498

Comparing different distributions between each other it's possible to notice that the tweaks made for the major overhaul didn't effect that much into the results, as it didn't do much of a difference in women's competition either. However when comparing the current distribution with old distribution, the men's standings are almost completely restructured. Only 3 athletes in top 15 were in the same position in both, and also in all three, ways of calculating the standings.

The only difference which the small tweaks had made for the top 15 was that Philipp Nawrath did lose couple positions. He was way more consistent and had better average performance as well with roughly equal amount of competitions with Vetle Sjaastad Christiansen and Martin Ponsiluoma but with the current system he ended up fairly clearly behind them. This just shows that both athletes mentioned who competed with Nawrath, had more finishes into the positions which got small buff during the season.

When comparing the new and the old system, the whole top 15 goes into a new order. Eric Perrot was dominant so he keeps his spot and Jacquelin on 5th place was confirmed but everyone else had a possibility to move around. With old system Sebastian Samuelsson would've beaten Sturla Holm Lägreid and Johan-Olav Botn. Nawrath would've climbed even one more position over Tommasso Giacomel, who did lose a lot of spots because of missing the last trimester. With old system even Ponsiluoma would've gained a position even still being behind Nawrath. Fillon Maillet and Campbell Wright was advancing while Johannes Dale-Skejvdal and Martin Uldal droppes positions above mentioned. Fabien Claude also beat Isak Frey to sneak into top 15 with old distribution. Even Lukas Hofer was almost gaining a spot but lost it with same points against Uldal because Uldal had better individual competitions results.

This is perfect showcase of what was probably tried to be achieved with change of the points distribution. In general it can be said that all the Norwegians have gained from the new system. The reason for that is not because they are Norwegian but because there is so high competition inside the national team and basically each everyone of them can win at given time. This creates situation where all of them probably will gain from the situation where more points are awarded for the top athletes. All of them had also less competitions compared to the competing athletes who would've positioned above them with the old distribution. In the case of Tomasso Giacomel this was also the case but with new points distribution and successful first two trimesters guaranteed him the 6th place over the 11th with the old system.

5.4.5 Men's top 15 with FIS related distributions compared to the current IBU distribution from season 25-26

Next take a look into a comparison of FIS related points distributions compared to the current IBU distribution.

IBU Current	Points	FIS Current	Points	FIS Old	Points
Perrot	1263	Perrot	1760	Perrot	1265
Lägreid	984	Samuelsson	1477	Lägreid	980
Botn	968	Lägreid	1388	Botn	936
Samuelsson	918	Botn	1382	Samuelsson	829
Jacquelin	876	Jacquelin	1357	Jacquelin	821
Giacomel	797	Ponsiluoma	1269	Giacomel	796
Christiansen	736	Nawrath	1246	Christiansen	644
Ponsiluoma	727	Fillon Maillet	1199	Dale-Skjevdal	619
Nawrath	716	Christiansen	1188	Nawrath	614
Dale-Skjevdal	697	Dale-Skjevdal	1120	Ponsiluoma	599
Fillon Maillet	689	Giacomel	1102	Uldal	545
Uldal	609	Wright	1089	Fillon Maillet	541
Wright	604	Claude F.	1017	Wright	471
Hofer	552	Hofer	1012	Hofer	398
Frey	538	Frey	1001	Frey	392

As seen in women's comparison, the current point distribution used by FIS Cross-Country World Cup changes drastically the positions of the athletes. There are only five athletes who remain on their position in the standings, Fabien Claude is climbing into top 15 in this version again and it's Martin Uldal who drops out. Athletes like Samuelsson, Ponsiluoma, Nawrath and Fillon Maillet gain the most as they were racing more actively over athletes like Lägreid, Botn, Christiansen ja Giacomel.

The old system however doesn't include too many changes. There are only two pairs who switch places, although Dale-Skjevdal pops us two places and Ponsiluoma drops those two places while Nawrath keeps his spot in the middle.

5.4.6 Closure of Men's top 15 comparison

To finalize the comparison on men's side, let's look a same table which was used at the end with women's case.

Name	Races	Ovr Pos	Alt Pos Avg	Avg Race Pos	Race Pos SD
Perrot	21	1	1	4.7	3.6
Lägreid	17	2	2.5	5.2	3.8
Botn	17	3	3.5	5.4	4.7
Samuelsson	21	4	3	10.3	14.9
Jacquelin	20	5	5	12.1	16.1
Giacomel	14	6	8.25	6.6	6.7
Christiansen	18	7	8	11.9	14.3
Ponsiluoma	21	8	8	13.1	11.0
Nawrath	19	9	7.25	10.0	5.0
Dale-Skjevdal	16	10	9.75	8.9	5.8
Fillon Maillet	21	11	10	13.6	7.0
Uldal	15	12	13	12.4	12.8
Wright	21	13	12.5	18.0	16.0
Hofer	20	14	14	16.8	9.3
Frey	21	15	15	18.3	12.2

In men's competition there's even more of alternation which can be seen through this table. There also more deviation on how many races athletes have been taken part but also the average positions on individual races and so the standard deviations vary slightly more. This is mainly due to the top athletes being even more consistent on getting to the top positions, which was seen in analysis earlier.

Since the men's results gave a lot more variation on how season end results would've looked like, it does effect on the average calculated from alternative distributions as well. In this case the averages shift quite a lot more, mainly due to the higher deviation of the races each athletes competed during the year. This example showcases even better how performing well, while you're competing, is possibly awarded over performing steadily in high level and participating in most if not all races throughout the year, when using this distribution. Looking into averages of other distributions in general those athletes, who had less races during the year, have average going down and athletes competing more have gained average up to a point where for example Sebastian Samuelsson have higher average than Johan-Olav Botn and there's couple other examples of this happening too. This is still not happening in all the cases and it's also good to remember that for this average there's also calculated even more top heavy distributions.

5.4.7 Wider comparison of the points distributions over the different seasons

Making a comparison from two separate competitions through same season can already give us a hint of how the changes in the point distribution can effect in the final standings. To make a more conclusive analysis it's possible to look into previous years and how the points distribution would've developed with different results.

When adding the sample size to include last 5 seasons, with women and men respectively, it's possible to make a more robust analysis of the effects of the different points distribution methods. Looking into the wider data set it quickly becomes clear that points distributions do have a major effect on the standings at the end of the season. First, lets look at the table which shows the average amount of effected athletes at the end of the season top 15 standings, when comparing the currently used point distribution in IBU World Cup to it's predecessors and FIS equivalents. The main looking points in this first look is to see how many athletes were affected in average and how many places did they move.

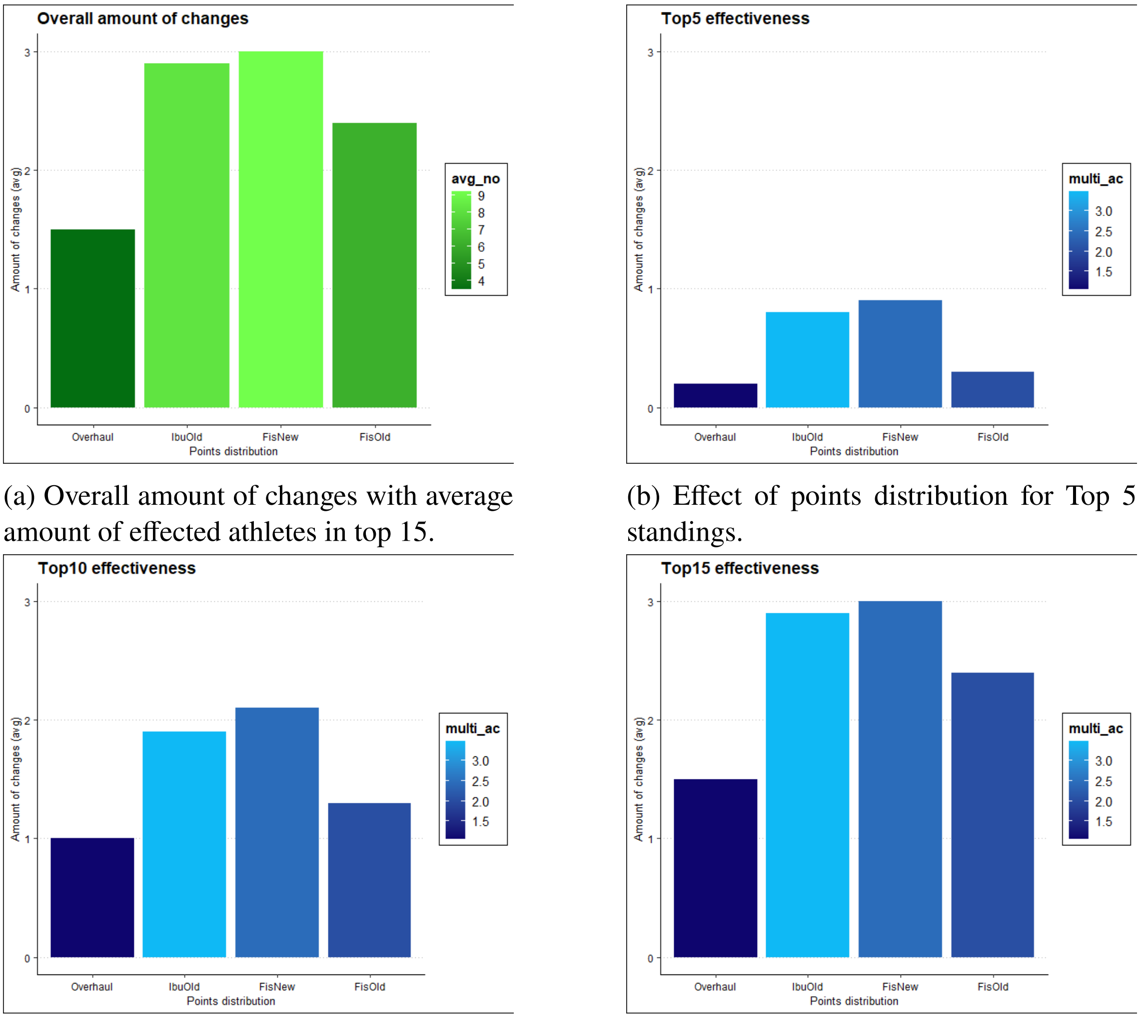
	IBU Overhaul	IBU Old	FIS New	FIS Old
Up 2 +	0,2	1,4	1,7	0,9
Up 1	1,6	2,4	3,4	1,9
Down 1	1,6	2,5	1,8	2,2
Down 2 +	0,1	1,1	1,7	0,8
New	0	0,7	0,8	0,4
Average	**3,5**	**8,1**	**9,4**	**6,2**

When comparing the current IBU distribution to it's predecessor (IBU Overhaul), the amount of athletes affected is low compared to the other systems. This is expected as the changes compared to the current system were the smallest. Normal change was between two athletes and the amount of changes were usually one or two where athletes switched their places.

The next closest to the current system was the old distribution of FIS Cross-Country World Cup which is also not a total surprise as it's the closest distribution to the other direction from the compared distribution. It did however start to effect quite a large part of the top 15 as it did affect over 6 athletes on every competition, which is 40% of top 15 standings moving around on average. This was also the only points distribution which did effect on the winner in one occasion.

Old IBU World Cup distribution and FIS World Cups new distribution did have even more shuffling in the top 15. The latter being the most drastic in it's differences ended up effecting over 9 athletes meaning, that over 60% of the top 15 standings were changed when calculating points with this system. The old distribution of IBU also did give drastically different standings with over 50% of the top 15 standings shuffled in average on each occasion.

Looking only for the amount of athletes changing their position in the standings isn't quite enough. In this case unless one athlete doesn't fall out from 15 and other replace him/her in there, one change effects always at least two athletes. It's also possible to calculate the amount changes based on this. The easiest change is that one athlete overtakes other but it's also possible that multiple athletes switch places into completely different order and all are effected by the different points distribution. So it's justifiable to look also how many changes did each distribution have and have many athletes each change on average did effect.



(a) Overall amount of changes with average amount of effected athletes in top 15.

(b) Effect of points distribution for Top 5 standings.

(c) Effect of points distribution for Top 10 standings.

(d) Effect of points distribution for Top 15 standings.

Figure 13: Set of graphs showing the effect of the points distribution compared to the current Ibu World Cup points distribution. In graph (a) the color gradient shows the average amount of athletes effected by the distribution and the rest of the graphs the color gradient shows a value of average amount of changes multiplied by standard deviation.

In the following graphs 13 this analysis has been done. There are total of 4 different graphs which showcases how different distributions compare to the current IBU World Cup distribution and how each distribution effects on different sections of top 15 standings. There are also color gradients which are telling us how many athletes distribution effected (in 13a) and rating value of distributions average amount of changes multiplied by it's standard deviation (in 13b, 13c and 13d). This graphs gives us more information about the differences of each distribution compared to the current IBU World Cup distribution. (NOT FINISHED)

It's also possible to compare where on average an athlete finishing on overall table would finish with different points distributions in play. In the last table let's look at how the average position changes compared to the current distribution over the results of 5 seasons. In the table there's also binary indication of if standard deviation of these numbers were over 1 (Yes, no bold) or over 2.25 (Yes, bold) which were decided to be a limits for high standard deviation in this case. Color coding was also added to indicate big differences with the current results. For example one change granted standard deviation of 0.316 and two 0.421. This mean that standard deviation of 1 did require already more than handful of changes which can be seen differentiating between distributions even the average position would remain close.

IBU New	IBU OH	High SD	IBU Old	High SD	FIS New	High SD	FIS Old	High SD
1	1		1		1		1.1	
2	2.0		2.3		2.7	Yes	1.9	
3	3.1		3.0		2.9		3.2	
4	3.9		4.4	Yes	4.3	Yes	3.8	
5	5.1		5.0	Yes	5.3	Yes	5.4	
6	5.9		6.5	Yes	6.5	**Yes**	6.1	
7	7.6	Yes	7.2		7.2	Yes	6.8	Yes
8	8.5		8.6	**Yes**	7.4	Yes	9.9	**Yes**
9	8.1		9.2	Yes	8.9	Yes	8.3	Yes
10	10.1		11.6	**Yes**	10.8	**Yes**	10.7	Yes
11	11.0		11.0	Yes	11.4	Yes	10.5	Yes
12	11.7		11.7	Yes	11.7	Yes	12.1	Yes
13	13.1		12.0	**Yes**	12.3	Yes	12.6	Yes
14	14.0		12.8	Yes	14.0	Yes	13.5	Yes
15	14.9		14.7	Yes	15.0	Yes	14.7	Yes

As seen on few earlier comparisons, overhauled version is fairly close and differences aren't too major. There's notable movement roughly where probably intended, at lower top 10s, where the latest tweak of distribution hit most heavily.

Other three distributions do have lot more differences in comparison. Old FIS distribution does give in general similar results in top 6 but after that especially the standard deviation of the results starts to change around. Biggest changes in average happens also in the same area with overhauled distribution, in lower top 10s. Old IBU distribution also shifts results with a fair amount compared to the currect distribution and especially the standard deviation is high, yet the average positions maintain fairly close to current system. Biggest changes on this comparison comes slightly lower in from 10th to 14th which can be explained by the distribution changes in top 10 but keeping the rest the same as before. This can be seen effecting on results around this area where single top finishes shuffle the positions around more with current system.

Finally the current FIS points distribution gives us the most deviation around the whole top 15 points table. On average the positions maintain fairly close to same they are in current distribution however which is interesting to point out. The reality is regardless that the averages come up from wider range of finishing positions when standard deviation is high. The distribution also effected heavier on wider range of positions, even more than old IBU distribution. One common thing was for all the distributions that it did effect title race rarely. The comparison did have only 10 different season competitions and adding up these would give us better understanding of how wide the distribution change will have effects for.

5.5 IBU Prize money distribution comparison

Next step is to compare prize money distribution. As the prize money is awarded to smaller amount of people it's expected that less athletes will be awarded by money prizes during the year. Total of 96 women and 95 men managed to finish in top 30 or be in a top 8 relay team at least once during the season 2025-2026. It's also not a big surprise that the Total Score winners, Jeanmonnot and Perrot, were able to snatch the biggest portion of the prizes. This obviously expected but in theory in a tight title race year it could be opposite. The main question is how much of the prize money will go towards the top and is it possible to determine a some kind of a limit, how many athletes can theoretically fund their professionalism via prize moneys earned during the year?

As IBU ain't publicly sharing prize moneys won by athletes, it won't be published in this report either for the respect of the athletes and IBU. Calculating the sum is still totally possible to anyone as all the information needed is publicly available. This comparison is made by using the percentages of how much of the total sum was each athlete was able to gather. This sum is gender based so the sum of percentages for women and men athletes respectively are hundred percent. This also gives better way to compare distribution between different sports with different amounts of prize money awarded throughout their seasons. When asking to the second question asked previously, it has to be taken into account that total prize money sums will differ. Looking into percentages can also reveal better the healthiness of the prize money distribution system than comparing straight sums of money.

As seen in graph (14) it follows fairly similar pattern than when comparing points in (12). It doesn't have as big of gaps which can be partially explained by calculation method but it also shows that even points distribution had bigger gaps, the prize money did even out between genders in fairly similar form.

One main interest of this comparison was to possibly see if there was much of mix in rankings of athletes with their earned points and prize money won throughout the season with different qualities. Interesting information gained while building this visualization during the last weekends of the season was that before the last weekend there were bigger mixes in between of athletes when compared points and prize moneys. After adding the final week and end of the season bonuses it all streamlined and only bigger outliers were the U23 cup winners who basically collected all the blue bib bonuses and won the U23 cup and therefore gained few positions in the prize money ranking over points ranking.

Even the end of the season bonuses did added some more top heaviness to the distribution, it was fairly subtle. The distribution of the prize money throughout the different bonuses is fairly interesting and also raises the question of the what is the reason for each of them. For example is the smaller discipline winner prize money's point to award the person who

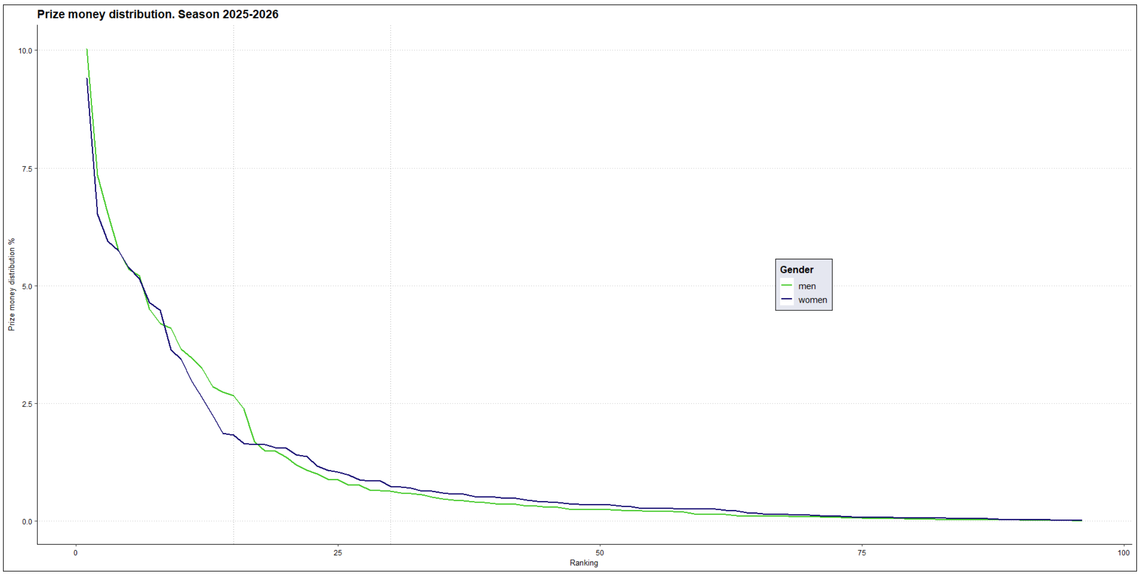


Figure 14: Comparison of prize money distribution on season 2025-2026. Comparison is made in percentages of total prize money distributed for the season 2025-2026 per gender.

ends up being the best in the discipline and this way share the prize money to more than one person or is it just a bonus for the Total Score winners if and when they are dominating the whole competition throughout the season? Is it also that with old points distribution method there was slightly more variance with the small discipline winners? Regardless of the reason, on this season both overall winners did also won three out of four smaller disciplines.

The answer to the second question of is it possible to determine a some kind of a limit, how many athletes can theoretically fund their professionalism via prize moneys, the first goal would be to set a limit for what is enough for an athlete to maintain her status as a professional athlete. Setting this kind of a limit is really difficult due the multiple different reasons. There are also multiple sources of funding which are possible for the athletes and usually the sponsor money is also more accessible for the top athletes than mid-level athletes. To give some kind of value to compare with cross-country in next chapter, around 60 women and 50 men were able to break a five figure prize money during season 2025-2026. This counts over one third of women athletes and over 25% of men athletes who participated at least one competition in IBU World Cup this season.

One major point to add is that IBU does have the second tier international competition circuit, IBU Cup, as well which part of the total prize money is distributed. With a quick glance over top athletes, it is confirmed that there are at least handful of athletes who are able to collect a five figure amount of prize money while only competing in IBU Cup. This is good information to take into account when comparing biathlon and cross-country prize money distribution in following chapter.

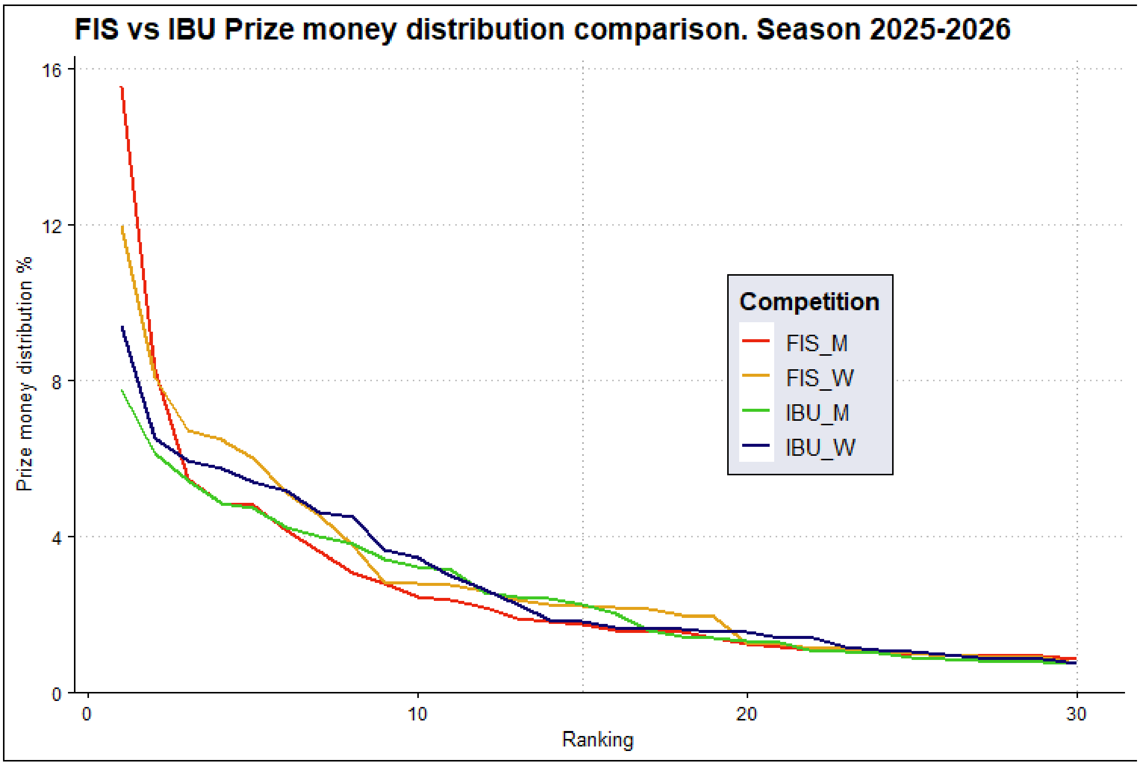


Figure 15: FIS World Cup vs IBU World Cup prize money distribution on top 30 most earned athletes in season 2025-2026

5.6 FIS vs IBU prize money comparison

Next the FIS cross-country World Cup prize money distribution is added to the graph to compare how the prize money is distributed within each sport. This comparison should give us more information about differences on distributions and can be used to compare the sports with each other on this aspect.

As mentioned earlier in IBU World Cup 96 women and 95 men were granted some amount of prize money over the season 2025-2026. In FIS cross-country World Cup 88 women and 120 men were granted some amount of prize money over the season. Since the FIS World cup information is taken straight from the source, the information of total amount of athletes participating into competitions throughout the year is not available. It is however interesting to note that in men's competition there was way more athletes who were capable on finishing in top 20 in individual competition or top 10 in team sprints to get awarded by prize money.

In the emphasis of trying to make comparison slightly more visually understandable, the graph is cut into two pieces. First graph visualizes the top 30 athletes based on prize money earned and the second graph will visualize the rest.

Looking at top 30 athletes (15) there are clear differences visible. First of all the top athletes of FIS World Cup are taking a bigger cut from the whole prize pool. Especially in the men's side the line is very top heavy. There's a good reason for that as one athlete, Johannes Hoesflot Klaebo, is almost unbeatable and therefore he receives almost the

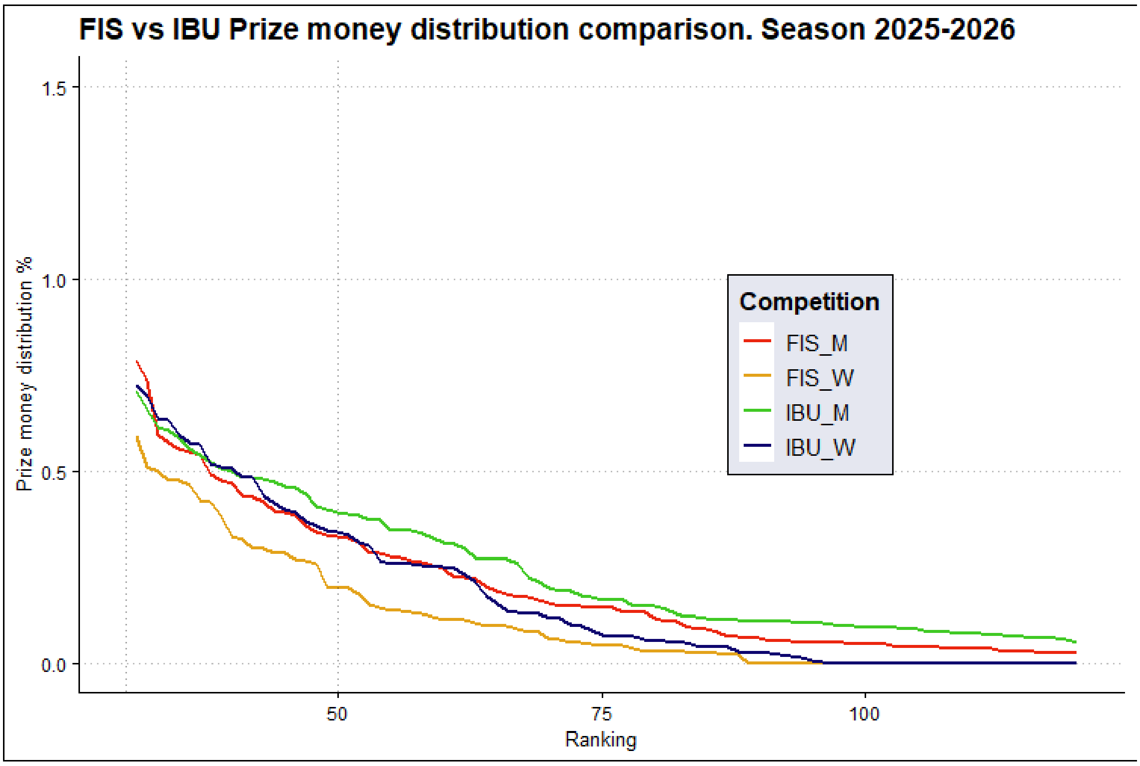


Figure 16: FIS World Cup vs IBU World Cup prize money distribution on the rest of the athletes who gained prize money in season 2025-2026

maximum amount of prize money one athlete can win per season. This has to be taken into account, as it was already earlier mentioned that when top athletes takes bigger cut, there's less to share more. This does show especially in this case.

Women's distribution is managing comparison slightly better until around 7th or 8th athlete where the drop is more drastic. After this the drop ends and athletes from 9th to 19th seem to get fairly similar cuts from the prize pool and for brief period they perform the best. Top 20 to top 30 the differences starts to level out and all the lines start to follow the same kind of path.

When looking the rest of the athletes from more closely there seems to be more differences but in grand scheme the differences aren't that big. It's good to mention that FIS World Cup's men athletes have bigger share for the whole time during this section compared to IBU's men but same can be said from IBU's women athletes in comparison to FIS's women. The major notice in this graph is that vastly wider range of men did manage to receive prize money in FIS World Cup than in other race formats. This is most likely due to higher competition divided into two different formats, sprint and distance athletes, and also two different skiing styles, classic and sprint, which was discussed about earlier. This is not as common in women's side which means that there are more women in the top who are highly competitive in sprint and distance competitions, with both styles.

When comparing the amounts of athletes who are capable of gathering a five figure prize money over the season, biathlon is performing way better compared to cross-country. in

FIS World Cup only around 35 athletes in women and under 40 athletes in men were able to collect a five figure money. In this light it could be said that IBU was able to distribute it's bigger prize money pool for wider group of athletes in bigger chunks, but the difference especially in the men's competition isn't that big. In women's competition the difference is much more noticeable and this really shows also that IBU's distribution does work as long as competition is equal enough in the wider range of athletes. The reality with FIS cross-country World Cup is that the overall prize money awarded is not big enough that the clearly wider range of athletes in men's competitions capable on finishing in top 20 wasn't enough to widen the range of athletes who were able to collect significant amount of prize money throughout the season.

One big point is also to count how many percent of athletes who participated in the World Cup competitions throughout the year managed to receive prize money. As the FIS World Cup data set was collected straight from the public source, there isn't collected statistics from this ratio. In IBU World cup 55,8% of women athletes and 50,8 % of men athletes manage grab their share of the prize money. This is fairly big part of all the athletes competing at least one during the season which can be seen as a positive result. The numbers for the FIS World Cup are not calculated due to only using the straight prize money share statistics from their official website.

6 Conclusion

Based on this analysis it's not yet completely possible to make definitive conclusions on either points distribution or prize money distribution effectivity. With further comparison it could be confirmed but changing the points distribution to award more points from top positions have made the overall competitions slightly less exciting at the end of the season as the differences between athletes are bigger. Of course it's good to mention that season 2025-2026 had Olympic Winter Games, which was the major goal for many top athletes but there was also some unfortunate events which effected the competition to overall victory in men's side. Having the smaller disciplines help with this challenge as these competitions are still tighter even at the end of the season. The latest changes on awarding athletes on the latter half of the top 10 with more points was a good addition.

There is a clear structure, a ladders to climb, for the athletes in the overall competition formats in IBU World Cup. There are 10 individual competitions where a whole pool of athletes are able to participate. There is a clear incentives for different levels. You have to be in top 60 best athletes in sprint to be able to compete in pursuit and add competitions in your calendar. It also gives you another chance to compete from points and even from more prize money. The next step is to get into top 40, in individual, sprint or pursuit, to earn points. Having top 30 to get awarded with prize money adds another step with something extra to compete from. in the end there's one more additional way to gather both points and prize money which is to perform in a high enough level to be able to participate in a mass start with top 30 athletes of given time but ideally athlete would like to be in top 25 in overall standings to quarantee this spot for every competition. Going closer to the top, athlete starts to compete from the individual podiums and wins, individual discipline wins and eventually overall Total Score win. The overall structure is very good and doesn't require major modifications. The relays give opportunities for the most of the athletes in the circuit to participate more competitions throughout the season.

In general IBU is also doing fairly good job with prize money distribution and proposals for the current system with the same pool of money are very limited. The amount of prize money given out has been rising year by year and as this is expected to be continuing it will be interesting to see how it will added to the current system. Based on this analysis some reasonable proposals could be made.

- Keeping the current prize money distribution for future year with IBU World Championships would be an absolute treat but likely not possible.
- If going back closer to the earlier years distribution, the emphasis could be trying to improve the tail end closer towards the this years distribution. Only after the tail is improved to this seasons standard, the winner's prize should be improved.
- Pursuit and relay leg bonuses were a great addition. The prize range is also good.
- Relays could award top 10 teams as in FIS team sprints.
- Total Score bonuses could reach up to top 15 athletes. Keeping the five figure bonus mininum is still more meaningful compared to the FIS World Cup comparison.
- Small disciplines could have top 3 athletes granted with a prize money instead of only winner, including the U23. This would mostly grant the top end but would give also some suprise athletes a possibility to gain a small extra at the end of the season.

Most of these proposals expect only marginal improvement on prize money pool. Keeping the season 2025-2026 level would require about 1,5 million euros added to the pool which most likely is not possible. In general the top athletes were able to collect a very hefty amount of money and therefore it would be good to aim to contribute most of the new money to the mid tiers at least for now. It's would be also nice to see if all the athletes who manage to end up on points, would be granted some prize money as well but as seen on this analysis, this wouldn't add up too many athletes to group of benefitters. It would most likely however improve the prize money gathered of the bottom half of those who already were granted some during this year which would improve the healthiness of the scene. Even with quick calculations this would also require quite a large sum of additional money.

In FIS cross-country World Cup the distribution is a lot more top heavy even disregarding the Kläbö dominance in men's overall competition. The improvements were done for it already in this year but it still doesn't feel good that third place finisher is getting less than 50% of the winner's prize money (read comparison in chapter 6). This part of the distribution got the major improvement compared to earlier years. In a way FIS is working with a completely different challenges in this regard as it would need to still focus on adding major improvement for 2-6 places in terms of prize money to make it feel much more worth while before even discussing of adding more placements who will earn prize money altogether. Overall bonuses at the end of the season for top 20 is relatively nice but the amounts after top 10 are so small that they are almost insignificant buff to what athletes on these positions have been already earning during the year. Even distributing money to wider group is better, it still feels like IBU is making better job with this part of the distribution as well. Distribution in FIS World Cup still needs a lot of improvement to be more healthy even the prize money pool wouldn't increase to the levels of IBU related events.

Viitteet

[1] FIS event calendar. FIS. <https://www.fis-ski.com/DB/general/calendar-results.html>

[2] FIS Cross-country schedule 2025-2026. FIS <https://www.fis-ski.com/DB/cross-country/calendar-results.html>

[3] FIS Prize money. FIS. <https://www.fis-ski.com/DB/v2/prize-money>

[4] Biathlon history. IBU. <https://www.biathlonworld.com/discover-biathlon/history/military-origins>

[5] Competition formats. IBU. <https://www.biathlonworld.com/discover-biathlon/how-it-works/competition-formats>

[6] Discover biathlon. IBU. <https://www.biathlonworld.com/discover-biathlon>

[7] How it works. IBU <https://www.biathlonworld.com/discover-biathlon/how-it-works/rankings-points>

[8] Prize money distribution 25-26. IBU. <https://assets.ctfassets.net/cz0vl36hcq0x/10xYsyJwytRCwMcj9
2026.pdf>

[9] 2025-2026 Standings. IBU. <https://www.biathlonworld.com/standings?DisciplineId=TS>

[10] World Cup Circuit. IBU. <https://www.biathlonworld.com/discover-biathlon/venues-series/world-cup-circuit>

[11] Cross-country skiing. Wikipedia. <https://en.wikipedia.org/wiki/Cross-country_skiing>

[12] Cross-country skiing (sport) Wikipedia. <https://en.wikipedia.org/wiki/Cross-country_skiing_(sport)>